2. Signal Representation and Synchronization

The physical representation of signals is dealt with in **layer 1 of the OSI model**.

In-depth knowledge of **signal theory** (cf. electrical engineering) would be required for an **extensive discussion** of these issues. We only present the fundamental ideas here.

2.1. The Physical Layer …

2.2. A Simplified Data Communications Model

2.3. Attenuation and Distortion

2.4. Frequency Domain Concepts

2.5. Relationship between Data Rate and Bandwidth

2.6. “Noise” (German: “Rauschen”)

2.7. Analog Transmission of Digital Data

2.8. Transmission of Digital Data with Puls Code Modulation

2.9. Communication requires Synchronization

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2.1. The Physical Layer …

The protocols of the physical layer specify:

– the **physical representation** of "0" and "1",

– the **duration** of a bit (i.e. the data rate),

– the **direction of the transmission** on the "channel",

a) **simplex**,
channel is only usable in one direction

b) **half-duplex**,
switching between sending and receiving

c) **full-duplex (=duplex)**
parallel transmission in both directions

– whether the transmission of bits is **serial** or **parallel**,

– whether the transmission is **synchronous** (in fixed time steps) or **asynchronous**,

– whether the channel is used **exclusively** or as a **multiplex channel**.

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Bit Errors in the Physical Layer

In any communication system, the signal received is different from the signal transmitted.
Usually, the most significant differences are caused by:

Attenuation (Dämpfung)

Gradual **loss off signal intensity** on the path from sender to receiver.

Distortion (Verzerrung)

The signal gets **deformed** while travelling from sender to receiver.

Noise (Rauschen)

The signal received includes **components NOT transmitted by the sender**.

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Example: A Bit Error resulting from Attenuation, Distortion and Noise

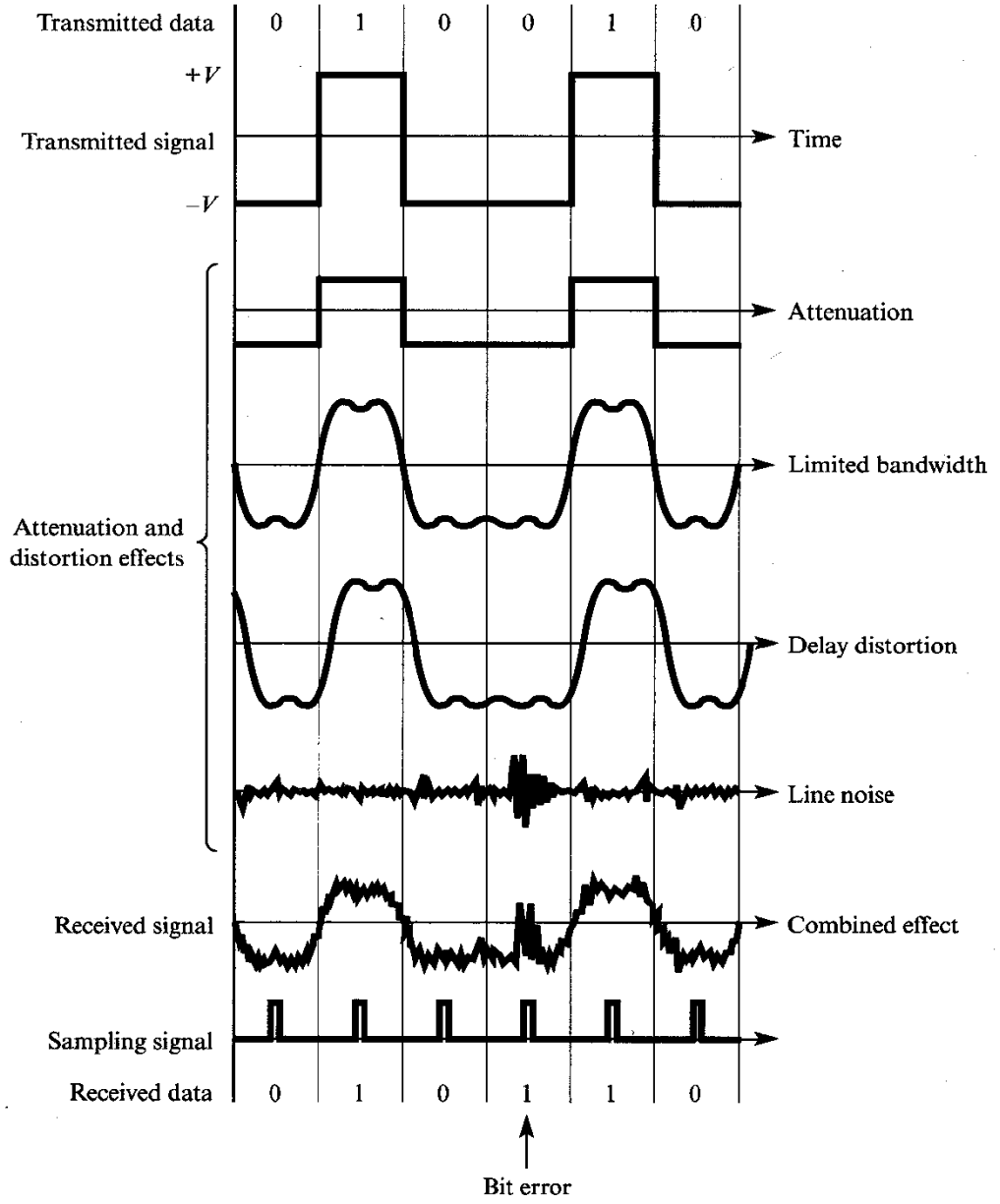
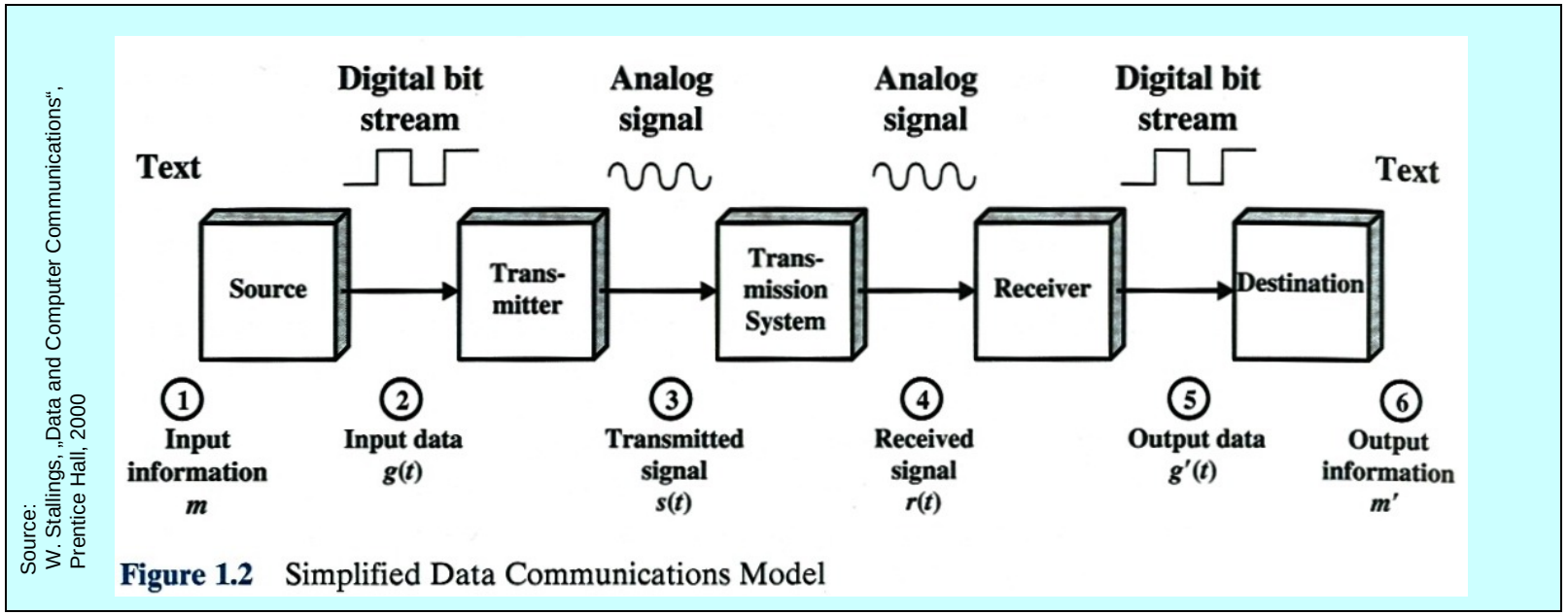


Figure 2.6
Sources of attenuation and distortion.

Source:
Fred Halsall
"Data Communications, Computer Networks and Open Systems", Addison-Wesley, 4th Edition, 1996

2.2. A Simplified Data Communications Model

Data are sent by generating a "**signal**" (a physical phenomenon representing "0"s and "1"s) and transmitting this signal over a medium.



Source: W. Stallings, „Data and Computer Communications", Prentice Hall, 2000

Figure 1.2 Simplified Data Communications Model

This transmitted/received **signal** is a **function of time**.
However, it can also be expressed as a **function of frequency**:

Time Domain View:

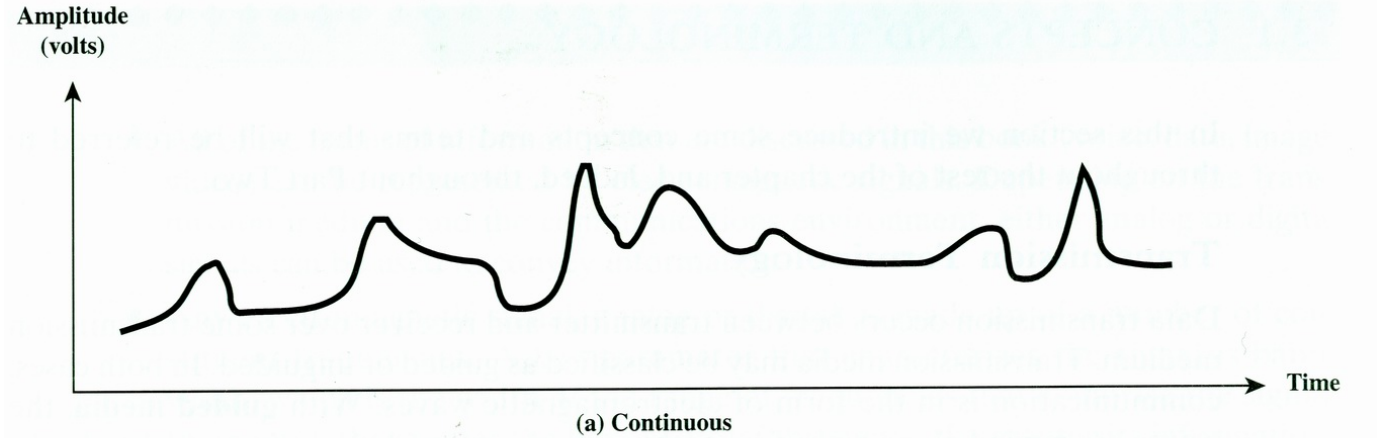
- Study the **signal intensity varying over time**
- The "natural" view.

Frequency Domain View:

- Study the **frequency components** of the signal,
- The often more important view.

Continuous and Discrete Signals

As a function of time, an electromagnetic signal may be either continuous or discrete.



Source:
W. Stallings, „Data and Computer Communications“,
Prentice Hall, 2000

Figure 3.1 Continuous and Discrete Signals

Periodic and Aperiodic Signals

In a **periodic signal**, the **same signal pattern repeats over time**.

Otherwise, the signal is **aperiodic.**

Mathematically, a signal is defined to be periodic if and only if:

$$s(t+T) = s(t) \qquad -\infty < t < +\infty$$

T: (constant) period of the signal

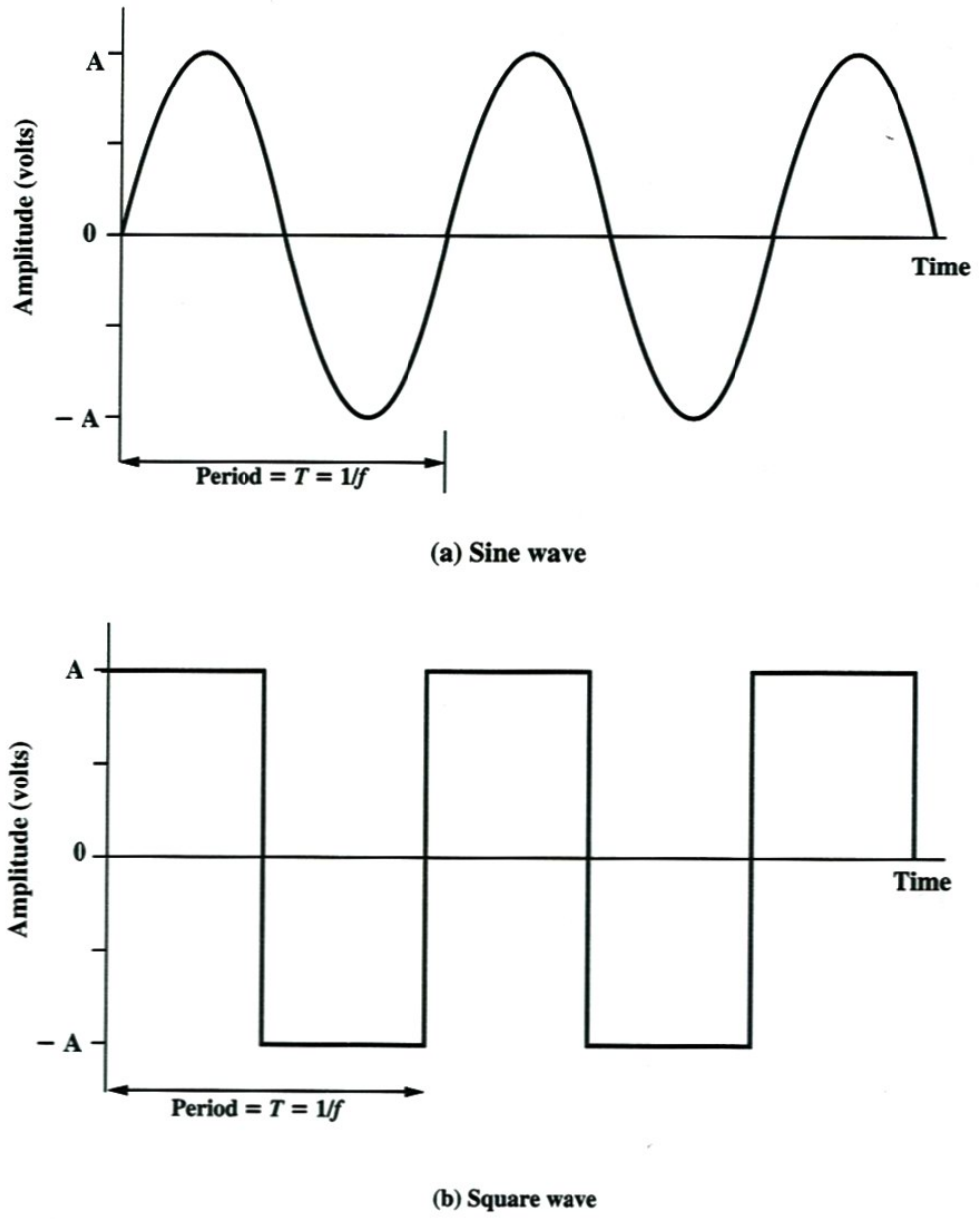
f = 1/T: frequency of the signal

The **frequency** is the rate (in cycles per second or Hertz [Hz]) at which the signal pattern repeats.

The **phase** is a measure of the relative position in time within a single period of a signal (as illustrated later).

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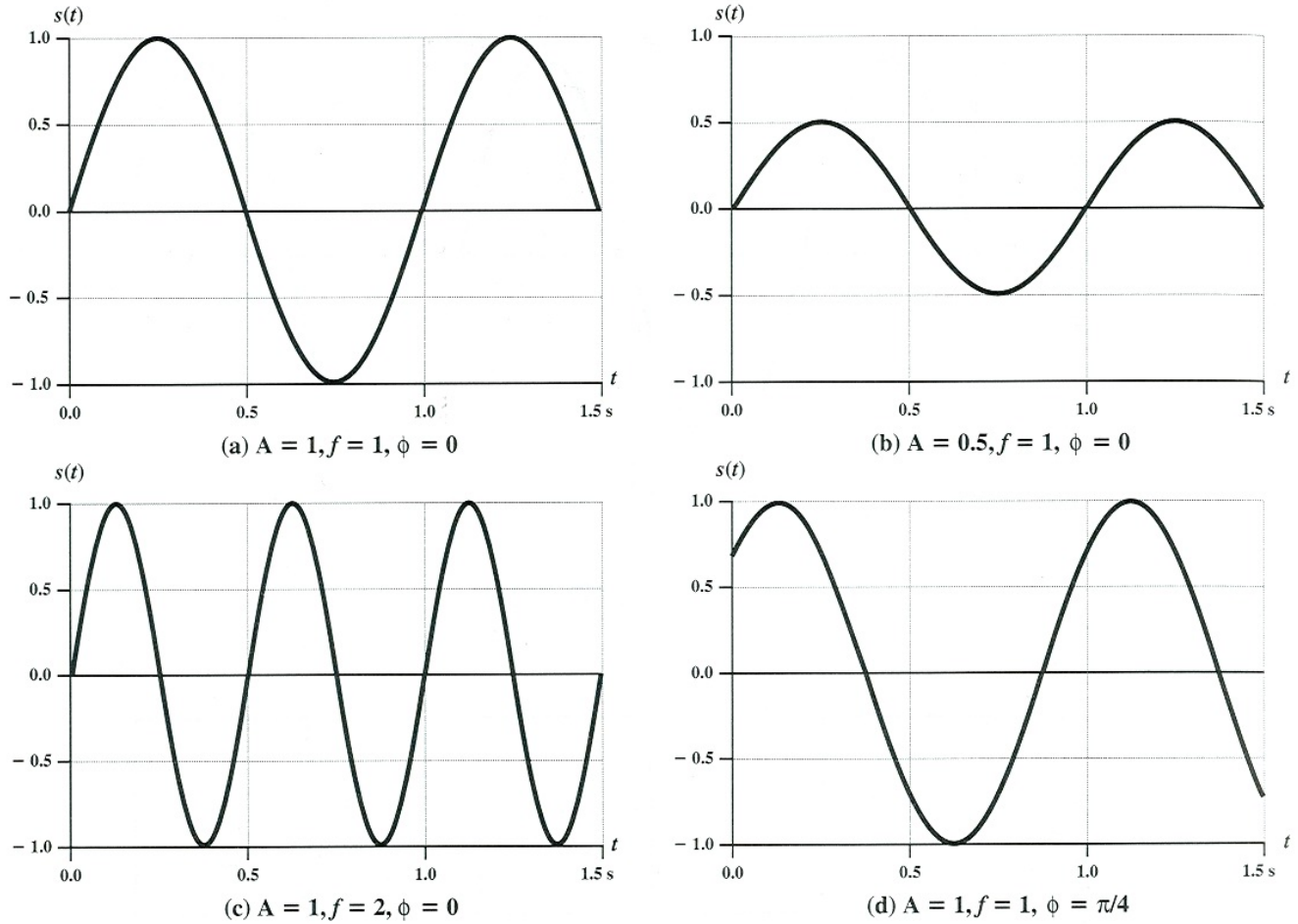
Example: Periodic Signals



Source:
W. Stallings, „Data and Computer Communications“, Prentice Hall, 2000

Figure 3.2 Periodic Signals

Example: Varying the Parameters of a Sine Wave



Source:
W. Stallings, „Data and Computer Communications",
Prentice Hall, 2000

Figure 3.3 $s(t) = A \sin(2\pi ft + \phi)$

With a change of scale, the same graphs can apply with horizontal axes representing space: The signal parameter varying over space (with fixed time).

The **wave length λ** is the distance occupied by a single cycle.
If the signal is traveling with speed v, then we find **λ = v •T**.

2.3. Attenuation and Distortion

Attenuation and **gain** (German: „Verstärkung“) usually are expressed in **decibel**:

$$N_{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

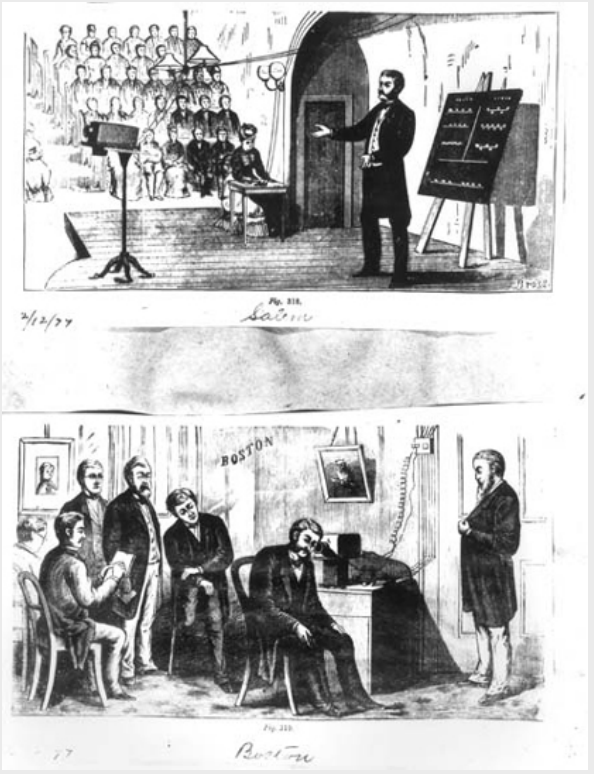
N_{dB}: number of decibels
P_1: input power level [W] unit Watt or mW
P_2: output power level [W]
$\log_{10}$: logarithm to the base 10, from now: “log” for short

The unit bel (and decibel) remind us of Alexander Graham Bell, * 1847 in Edinburgh, Scotland, † 1922 close to Baddeck, New Scotland, the inventor of the telephone (with Ph. Reis) and founder of AT&T.

The unit decibel (dB) is extensively used in electrical engineering and acoustics to express the ratio of two parameters with a logarithmic scale.

Bel and decibel obviously have no dimension. However they are used for expressing absolute values by referring to reference values.

Alexander Graham Bell demonstrating the telephone, 1877.



Source: http://www.att.com/history/inventing.html

Attenuation (2)

"The **decibel** (abbreviated dB) is a **logarithmic unit** that can be used to **measure ratios** of virtually anything."

W. Tomasi, „Electronic Communications Systems – Fundamentals Through Advanced, 5th Edition, Prentice Hall, 2004, p. 2

Example:

If a signal of a power level of 10 mW is inserted onto a transmission line and the measured power some distance away is 5 mW, the loss may be expressed as:

$$N_{dB} = 10 \log (5/10) \approx 10 (-0{,}3) = - 3 \text{ dB}$$

Table 3.4 Decibel Values

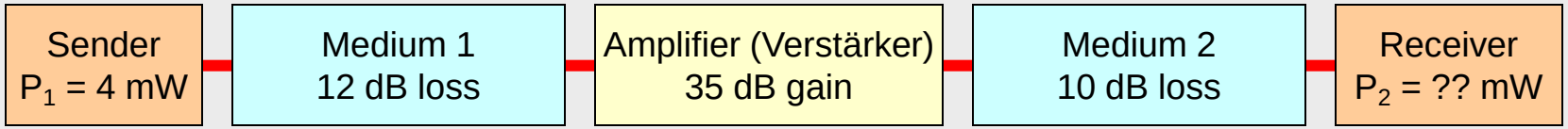
Power Ratio	dB	Power Ratio	dB
10^1	10	10^{-1}	–10
10^2	20	10^{-2}	–20
10^3	30	10^{-3}	–30
10^4	40	10^{-4}	–40
10^5	50	10^{-5}	–50
10^6	60	10^{-6}	–60

Source: W. Stallings, „Data and Computer Communications", Prentice Hall, 2000

Loss/Gain in Cascaded Transmission Paths

Describing the signal strength in dB allows for a simple calculation of what happens along a transmission path.

Example: Loss/Gain in a Cascaded Transmission Path



As total loss/gain we find: -12 + 35 – 10 = 35 – 22 = 13

Thus, the signal strength at the receiver may be calculated as:

$$13 = 10 \log (P_2 / 4 \text{ mW})$$

$$\mathbf{P_2 = 4 \bullet 10^{1,3} \text{ mW} = 79,8 \text{ mW}}$$

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Attenuation of Typical Guided Media

In the wired world, the attenuation is usually expressed as **attenuation per 100 m** or per km. Some examples:

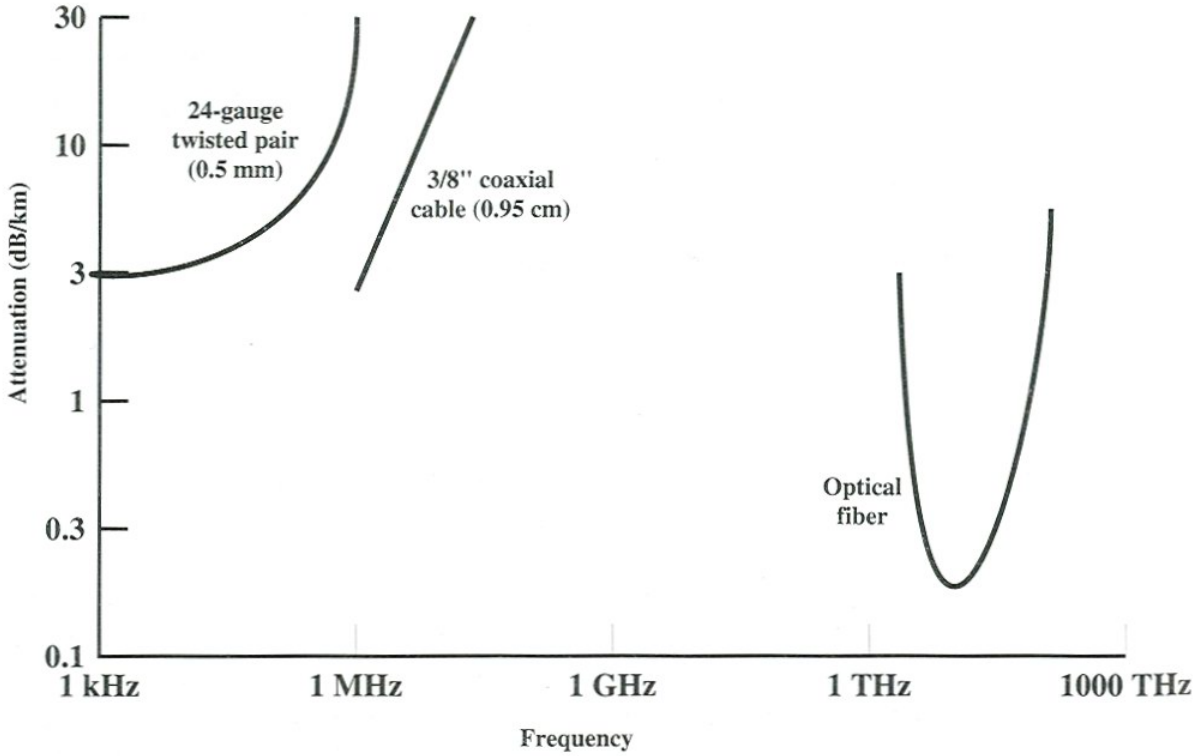


Figure 4.3 Attentuation of Typical Guided Media

Source: W. Stallings, „Data and Computer Communications“, Prentice Hall, 2000

Attenuation per 100 m in dB						
Frequency (MHz)	1	4	16	25	100	300
UTP Category 3	2,6	5,6	13,1	-	-	-
UTP Category 5	2,0	4,1	8,2	10,4	22,0	-
STP 150 Ohm	1,1	2,2	4,4	6,2	12,3	21,4

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Radio Links: “Free Space Path Loss“

Wireless communication links often are described by the “Free Space Path Loss“ model which is a model for the power degradation of electromagnetic waves transmitted to “**free space”** (no absorption, no reflexion).

Mathematically, the “Free Space Path Loss“ is described as:

$$L = \left(\frac{4\pi d}{\lambda}\right)^2 \quad \text{or} \quad L = \left(\frac{4\pi f d}{c}\right)^2$$

L: free-space path loss (unitless)
d: distance (meters)
f: frequency (hertz)
λ: wavelength (meters)
c: velocity of light in free space (3 x 10^8 m/s)

Expressed in dB, we find:

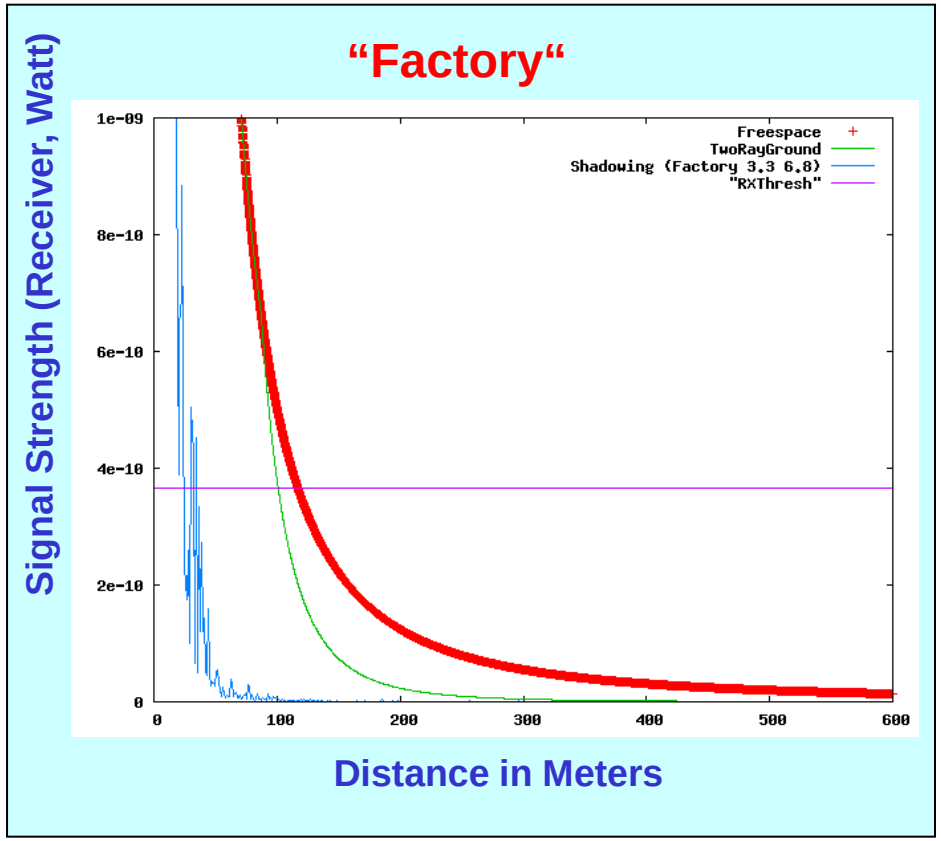
$$L(\text{dB}) = 10\log\left(\frac{4\pi f d}{c}\right)^2 \text{ dB} \quad \text{or} \quad L(\text{dB}) = 20\log\left(\frac{4\pi f d}{c}\right) \text{ dB}$$

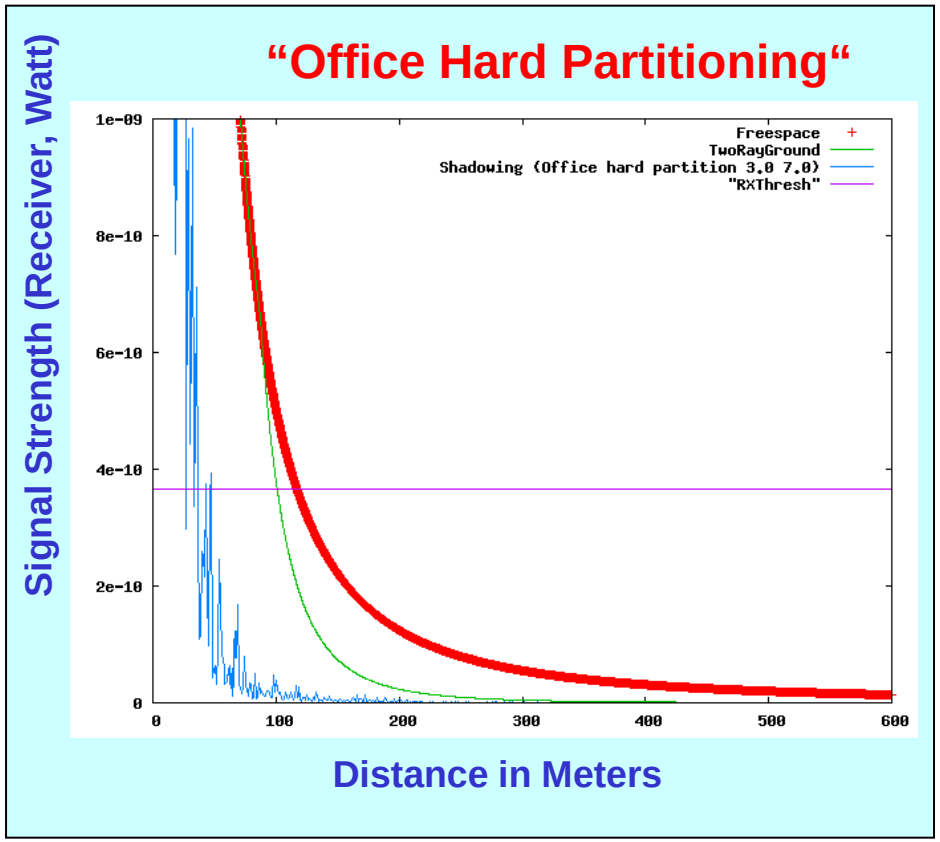
Example: d = 50 km, f = 6 GHz

$$L(\text{dB}) = 20\log\left(\frac{4\pi \cdot 6\bullet 10^9 \bullet 50 \bullet 10^3}{3\bullet 10^8}\right) \text{ dB} \approx 142\,\text{dB}$$

Reality Check: “Free Space Path Loss“ ?

In real life, we sometimes find drastic deviations from the Free Space Path Loss model. In the framework of a “Praktikum” in our group, one of our students presented the following **graphs for Wireless LAN**.



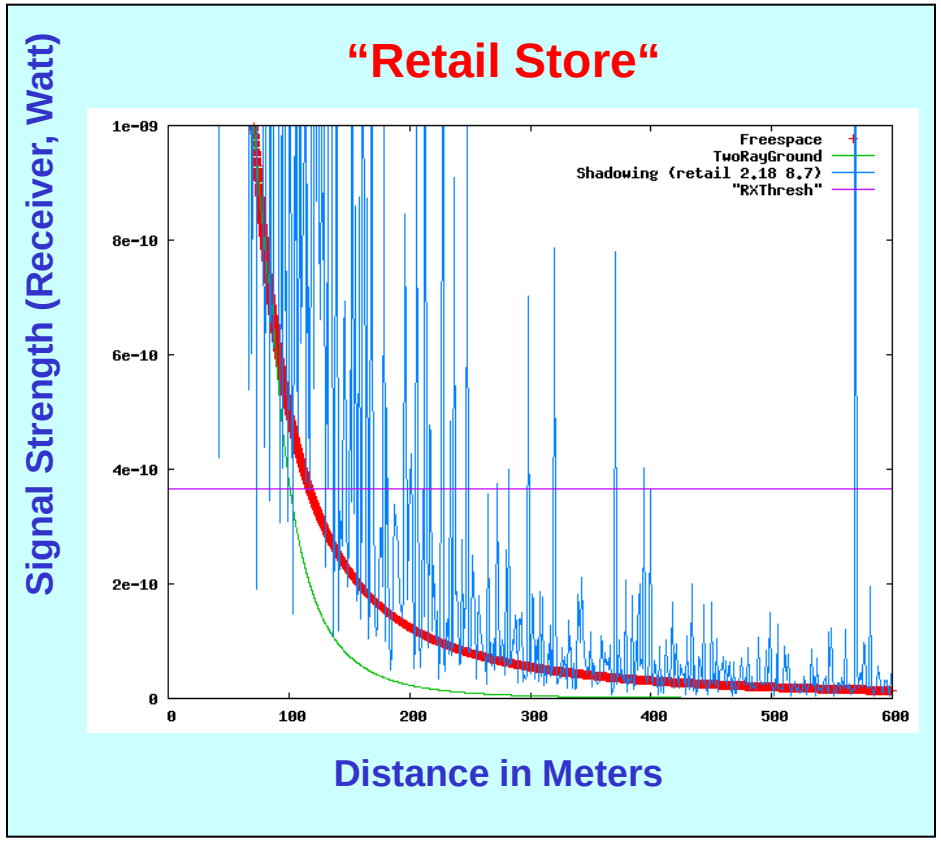


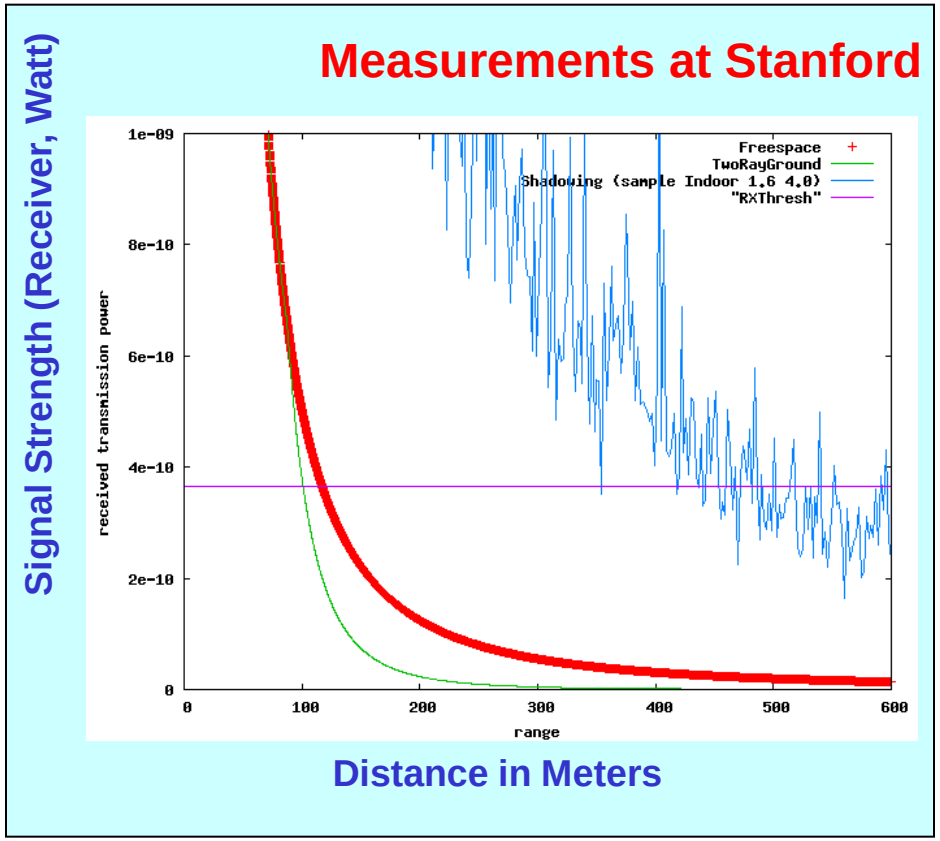
Freespace:	calculated with Free Space Path Loss – model
TwoRayGround:	more realistic model for relatively large distance
Shadowing:	model including “fading“ and attenuation
RXThresh:	minimum signal strength required at the receiver

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Reality Check: “Free Space Path Loss“ ? (2)

For all scenarios, the transmitter power was chosen such that a distance of 100 m was reached with the “TwoRayGround Model“.





Freespace: calculated with Free Space Path Loss – model
TwoRayGround: more realistic model for relatively large distance
Shadowing: model including “fading“ and attenuation
RXThresh: minimum signal strength required at the receiver

dBm - values

Often **1 mW** is used as reference value and defined as **0 dBm**.

It follows:

$$\text{Power}_{\text{dBm}} = 10 \log \frac{\text{Power}_{\text{W}}}{0{,}001\ \text{W}}$$

Examples:

1000 W	60 dBm
1 W	30 dBm
1 mW	0 dBm

Power in W	Power in dBm
0,001	0 dBm
0,002	3 dBm
0,01	10 dBm
0,1	20 dBm
1	30 dBm
10	40 dBm
100	50 dBm
1.000	60 dBm
10.000	70 dBm
100.000	80 dBm

Power in mW	Power in dBm
1	0 dBm
0,5	- 3 dBm
0,1	- 10 dBm
0,01	- 20 dBm
0.001	- 30 dBm
0,0001	- 40 dBm
0,00001	- 50 dBm
0,000001	- 60 dBm
0,0000001	- 70 dBm
0,00000001	- 80 dBm

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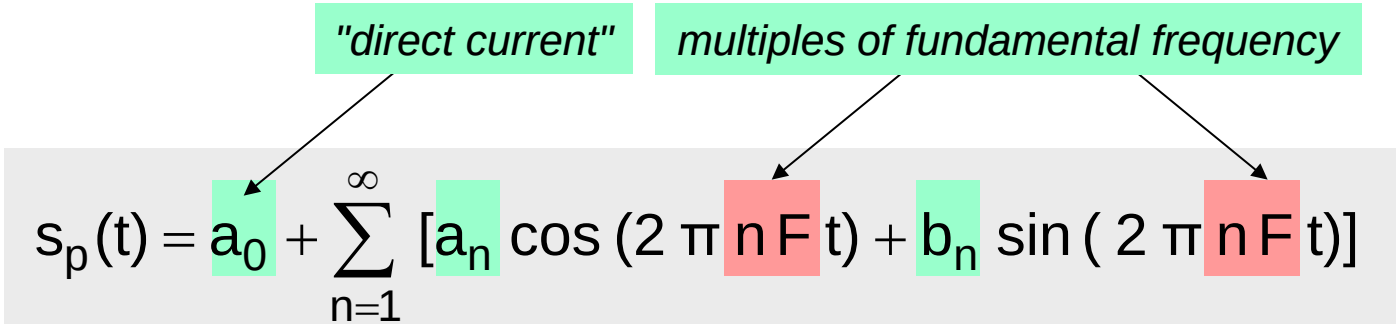
2.4. Frequency Domain Concepts

From mathematics and signal theory we know:

A periodic function (with period T) may be described as a **sum** of

- **weighted sine functions** and
- **weighted cosine functions**

More precisely: The Fourier Series



F = 1/T is the **fundamental frequency**.

More about this: Fourier analysis ("harmonic analysis")

Periodic Signals and their Fourier Series

Source:
W. Stallings, „Data and Computer Communications“,
Prentice Hall, 2000

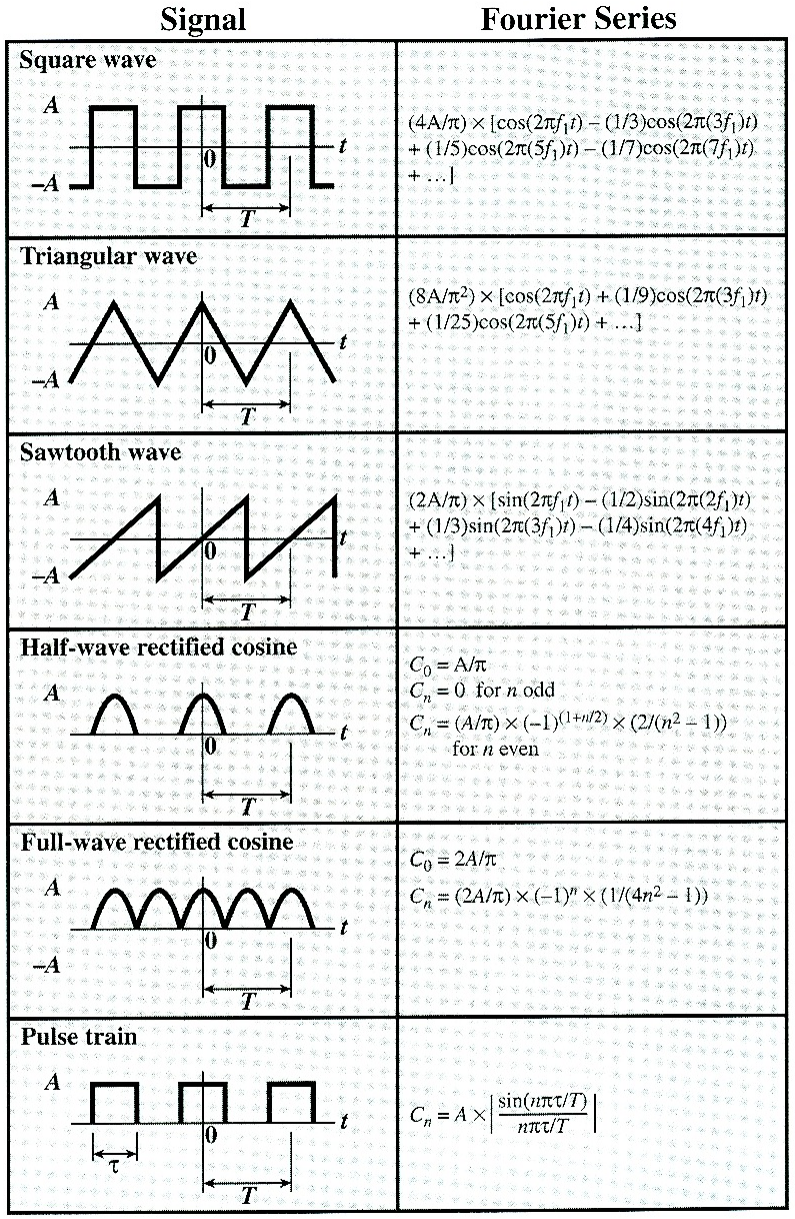


Figure 3.15 Some Common Periodic Signals and Their Fourier Series

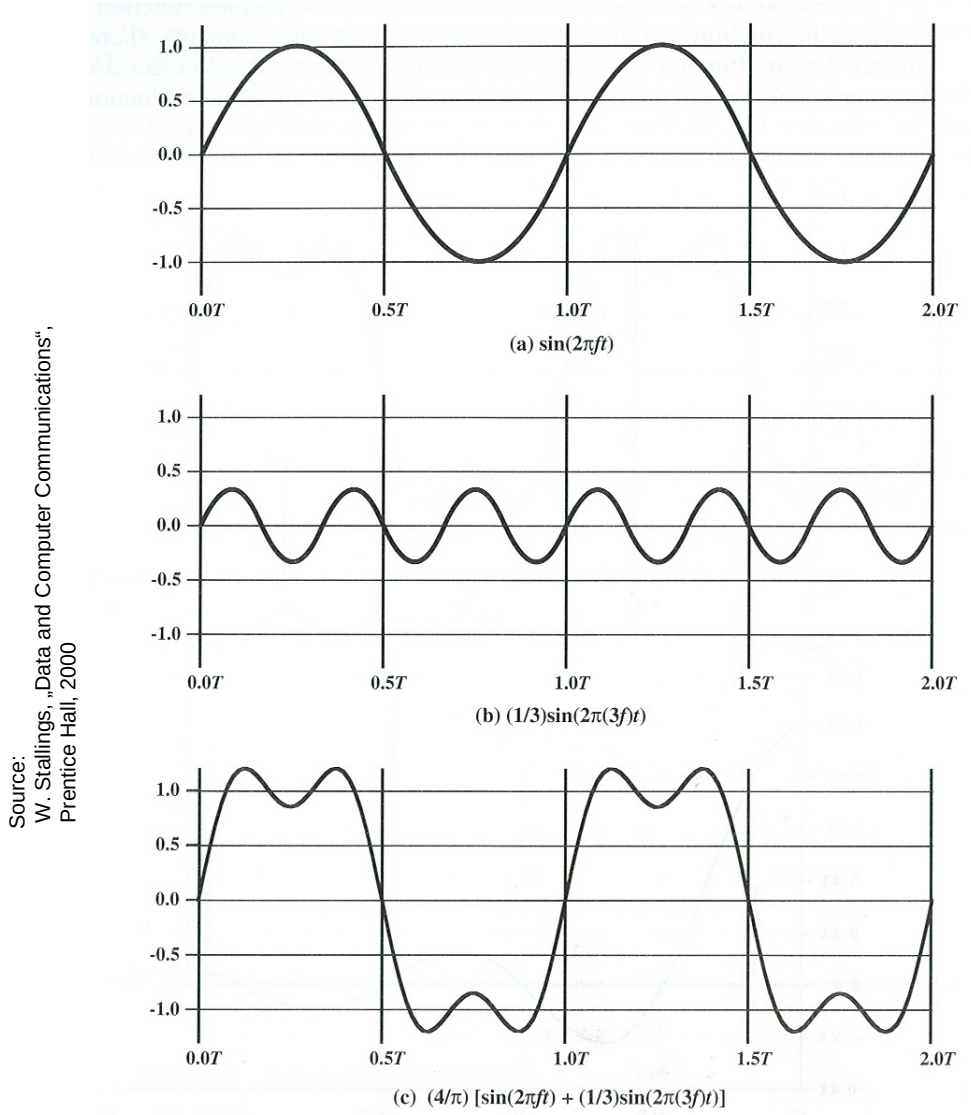
Example: Addition of Frequency Components

Real-life electromagnetic signals include many frequencies.

As an example, we study the **addition of two sine waves with different frequencies**.

Note:

The frequency of signal (b) is an integer multiple of the frequency of signal (a).



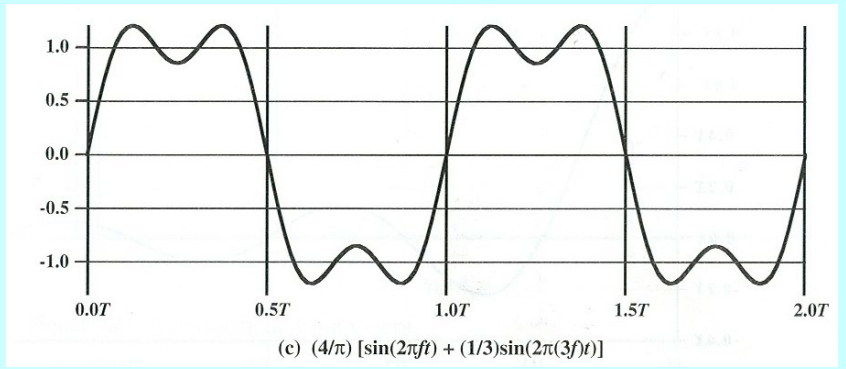
Source: W. Stallings, „Data and Computer Communications", Prentice Hall, 2000

Figure 3.4 Addition of Frequency Components ($T = 1/f$)

Signal Representation in Time Domain and Frequency Domain

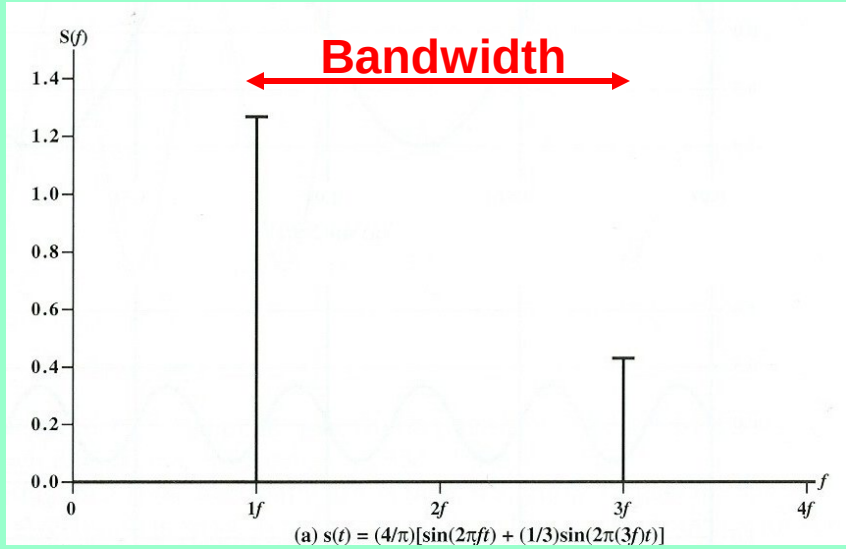
Time Domain Representation:
Signal amplitude varying over time.

Signal represented in the Time Domain



Frequency Domain Representation:
Peak amplitudes of the signal components.

The same Signal in the Frequency Domain



Source: W. Stallings, „Data and Computer Communications“, Prentice Hall, 2000

The “**spectrum**” of a signal is the range of frequencies it contains.

The “**absolute bandwidth**” is the width of the spectrum. Here: 2 f

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Fourier Transform

The spectrum of an **aperiodic** function consists of a **continuum of frequencies**:
The "**Fourier Transform**" [from the time domain] to the frequency domain and the reverse transformation are given by:

$$S(f) = \int_{-\infty}^{+\infty} s(t)\, e^{-j2\pi f t}\, dt \qquad \text{and} \qquad s(t) = \int_{-\infty}^{+\infty} S(f)\, e^{j2\pi f t}\, df$$

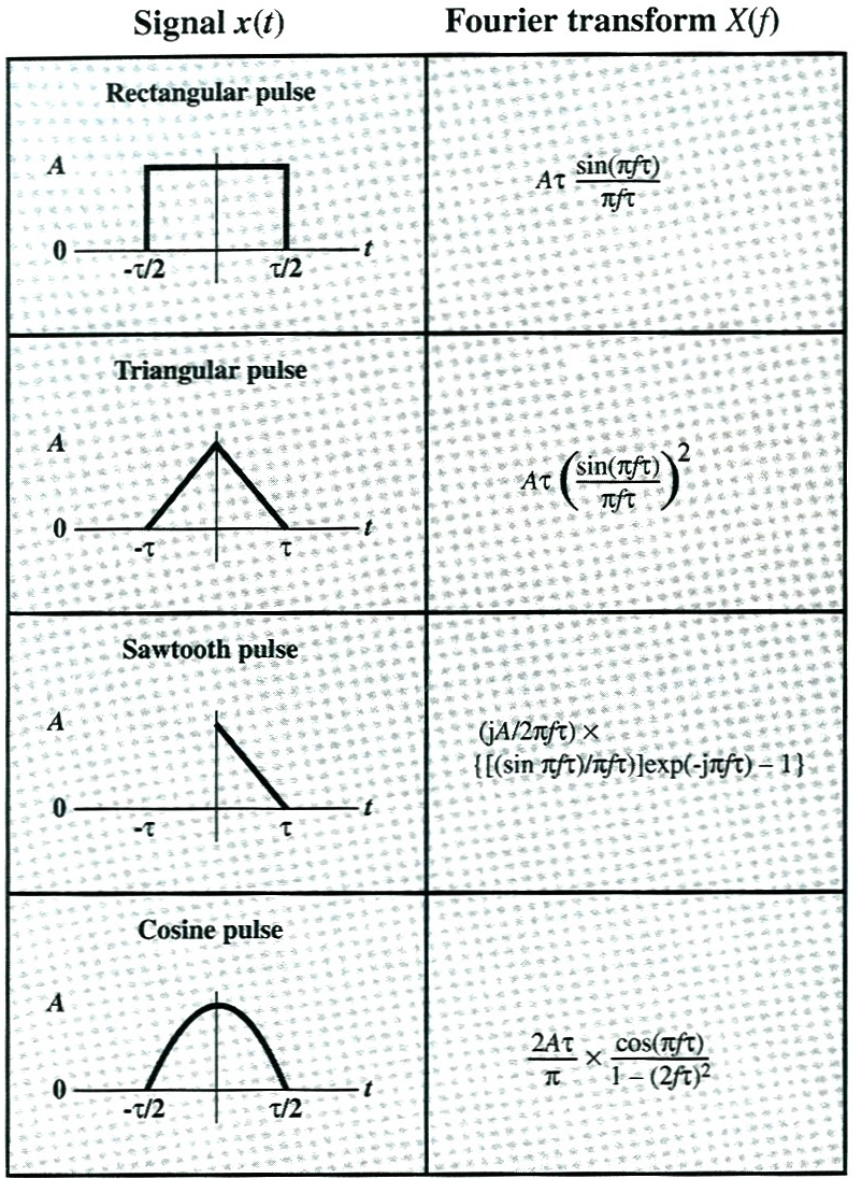
The "**Fourier spectrum**" S(f) basically is an infinite sum of sine and cosine waves with frequencies varying in a continuum.

Note:

- Being a transform, no information is created or lost in the process. So the original signal can be recovered from knowing the Fourier transform, and vice versa.
- Usually, we are only interested in lower and upper band limits ("Grenzfrequenzen").
- „j" is the imaginary unit, i.e. $j^2 = -1$
- Remember the **Euler Formula**:

$$e^{j2\pi ft} = \cos(2\pi ft) + j\sin(2\pi ft)$$

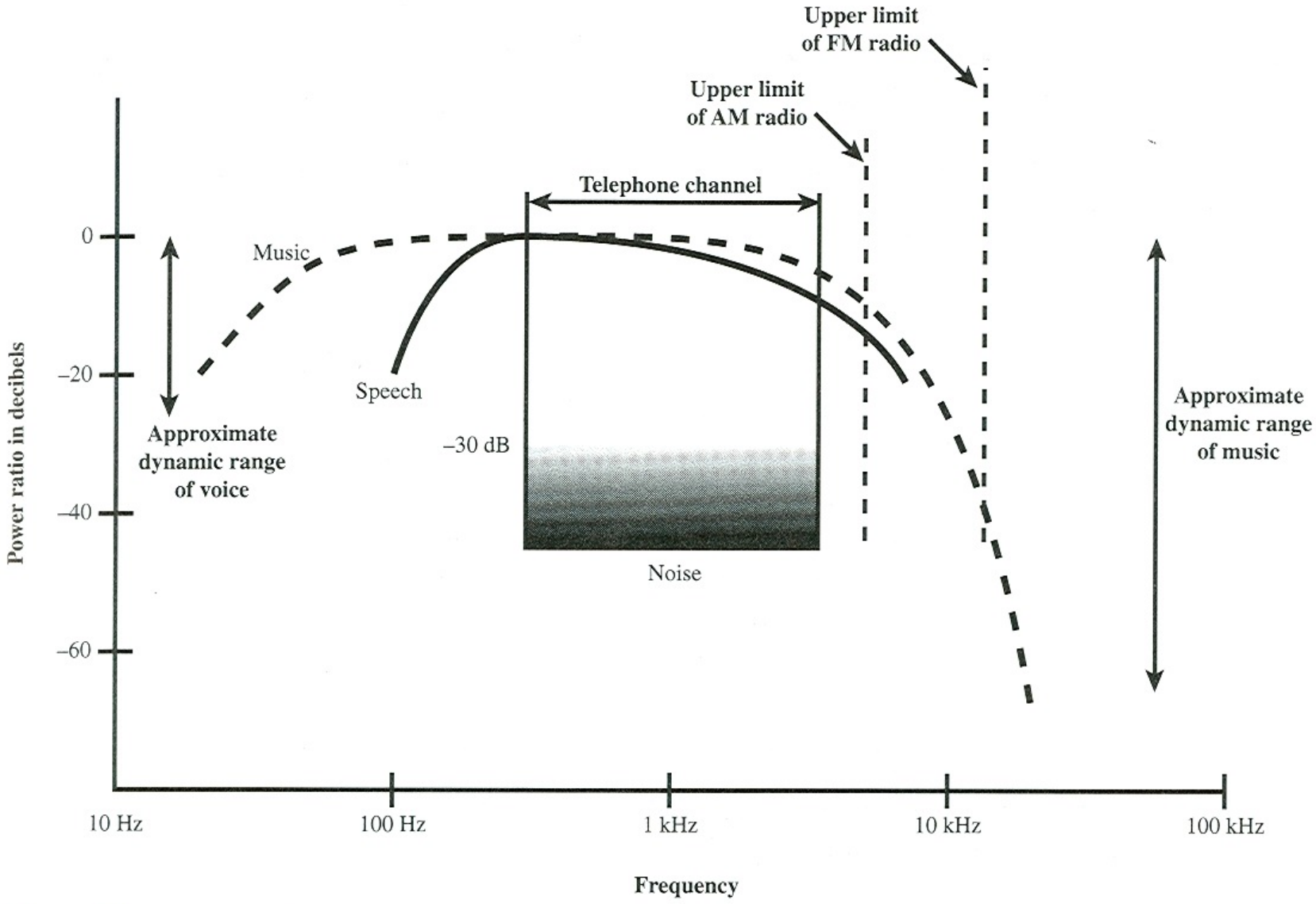
Aperiodic Signals and their Fourier Spectrum



Source:
W. Stallings, „Data and Computer Communications“,
Prentice Hall, 2000

Figure 3.16 Some Common Aperiodic Signals and Their Fourier Transforms

Acoustic Spectrum of Speech and Music



Source:
W. Stallings, „Data and Computer Communications“,
Prentice Hall, 2000

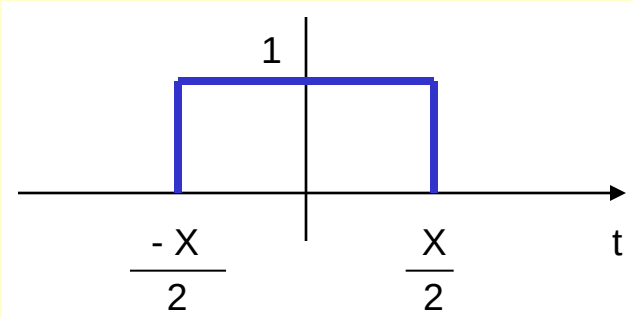
Figure 3.9 Acoustic Spectrum of Speech and Music [CARN95]

Example: Signal Representation in Time Domain and Frequency Domain

As an example for a signal with a continuous spectrum, we study the **single square pulse** which has the values

1 between –X/2 and X/2
0 else

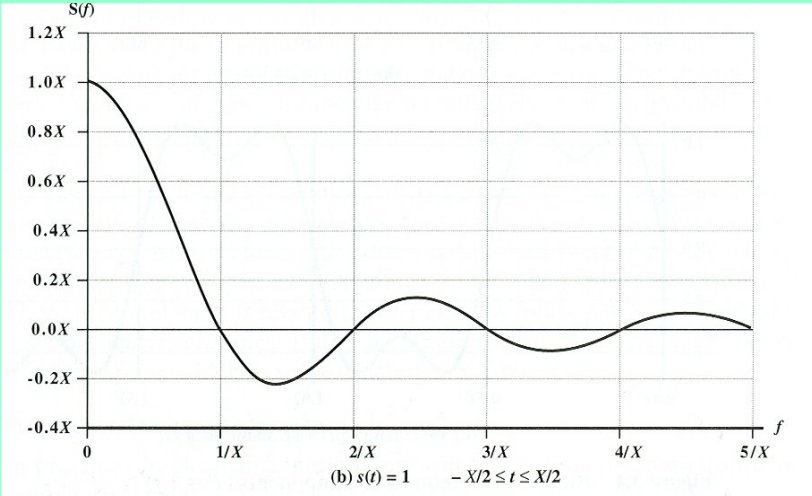
A Signal represented in the Time Domain



Obviously, this signal has nonzero components with extremely large frequencies.

However, **most of the energy is in a narrow band of frequencies**. This band is called "**effective bandwidth**" or just "**bandwidth**".

The same Signal in the Frequency Domain

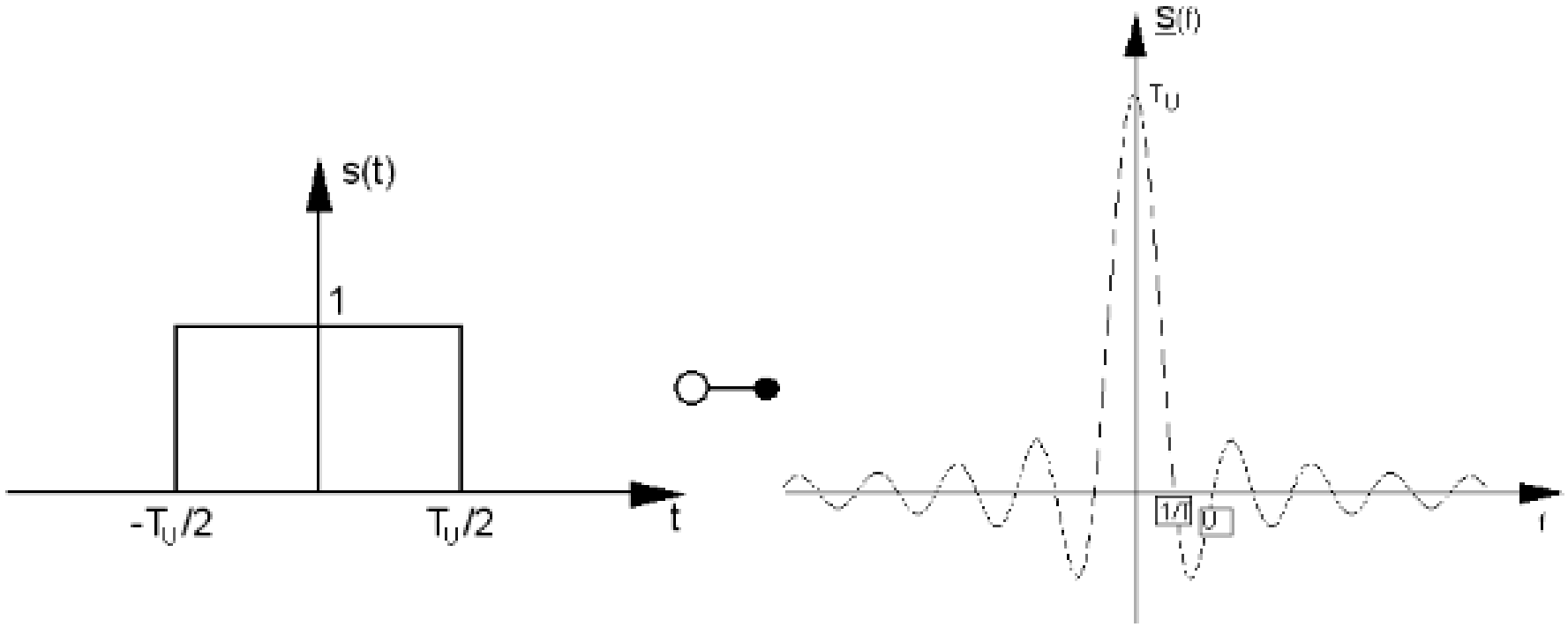


Source: W. Stallings, „Data and Computer Communications", Prentice Hall, 2000

Quote from Stallings,
„Data and Computer Communications":

In fact, the function S(f) for this case is **symmetric around f = 0**, and so has values for negative frequencies. The presence of negative frequencies is a mathematical artifact whose explanation is beyond the scope of this book.

Single Square Pulse and its Spectrum (“Si function“)



2.5. Relationship between Data Rate and Bandwidth

The "**effective bandwidth**" is the band within which ***most* of the signal energy** is concentrated.

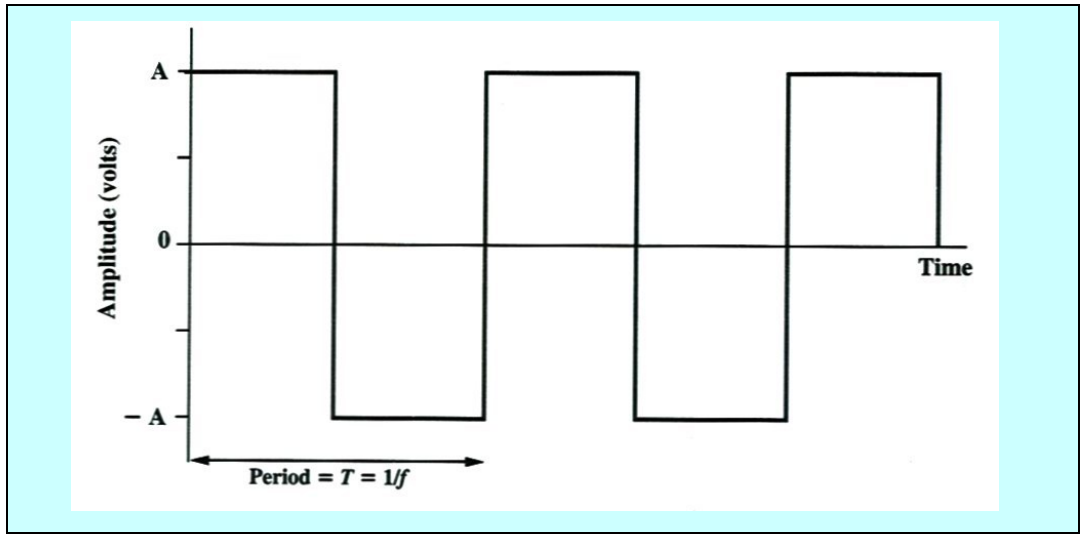
Real senders, transmission media and receivers can only accommodate a limited band of frequencies. This limits the data rate which can be carried on the transmission medium.

Now consider an example where the binary stream **1010...** is represented by

1 **positive pulse** und,

0 **negative pulse**.

Then we find the square wave:



Question:

What is the spectrum of this signal?

Frequency Components of the “Square Wave“

It can be shown:

The square wave with amplitudes A and –A can be expressed as:

$$s(t) = A \bullet \frac{4}{\pi} \bullet \sum_{k \text{ odd}, k=1}^{\infty} \frac{\sin(2\pi k f t)}{k}$$

This waveform has an infinite number of frequency components, i.e. **unlimited absolute bandwidth**.

However, the **amplitudes of high frequency components are very small.**

The graph shows the effect of dropping high frequency components:

A “distorted” square wave.

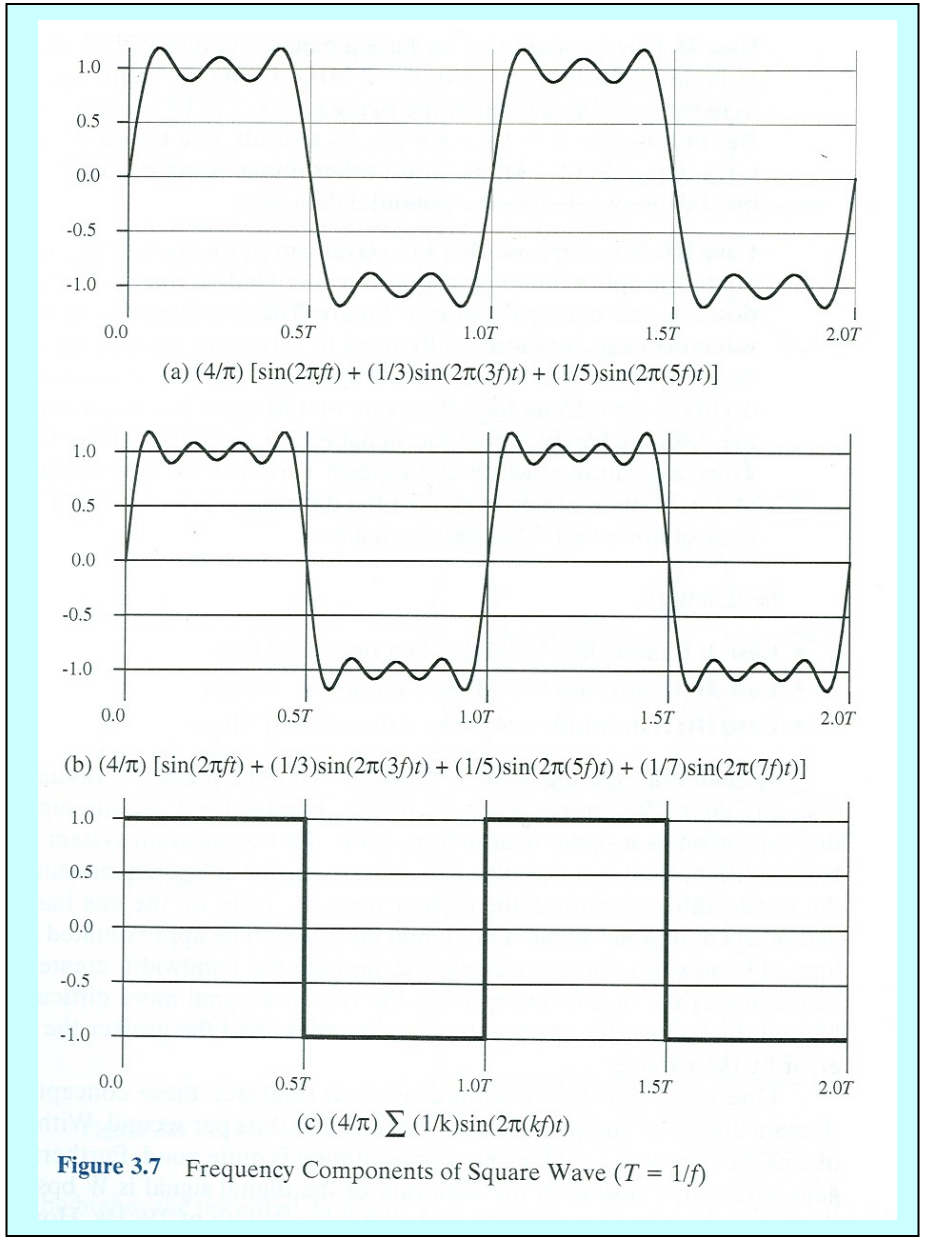


Figure 3.7 Frequency Components of Square Wave ($T = 1/f$)

Source: W. Stallings, „Data and Computer Communications“, Prentice Hall, 2000

Effects of the Channel Bandwidth on a Digital Signal

Finally, we study the transmission of a 2,000 bit/s data signal over channels with different bandwidths.

As a **rule of thumb** we learn:

For a **W bit/s digital signal a very good representation may be achieved with a 2W Hz channel**.

However, unless noise is very severe, the bit pattern may be recovered with much less bandwidth than this.

On the other hand: The more limited the bandwidth, the larger the distortion, … and the greater the potential for errors at the receiver!

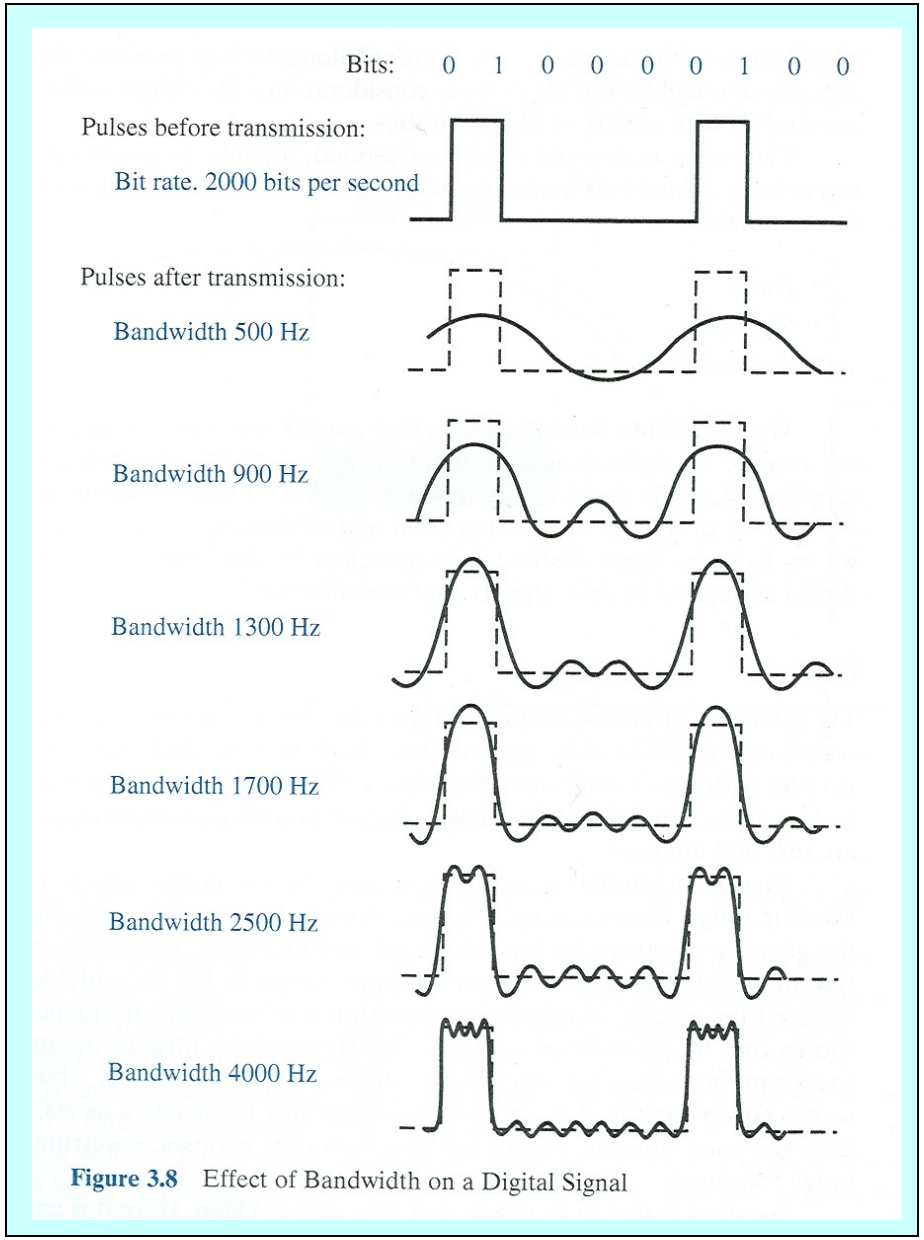


Figure 3.8 Effect of Bandwidth on a Digital Signal

Source: W. Stallings, "Data and Computer Communications", Prentice Hall, 2000

Maximum Channel Capacity according to Nyquist

In theory, digital signals have an **infinite spectrum**.

In **real life,** we always find components with "**lowpass characteristics**", i.e. components cutting off all high-frequency components of the signal.

Harry Nyquist has shown for noise free channels:

$$C = 2\,B$$

C: max. channel capacity in bit/s

B: channel bandwidth in Hertz

Example: "Classical" telephone channel

- **Bandwidth 3.100 Hz**, i.e.
- **max. 6.200 bit/s**

In reality, we find higher data rates.

Was Nyquist wrong ?

Instead of signals with two levels (each signal element represents one bit), we may use **"multilevel signaling".**

Example: Using **4 voltage levels** (instead of 2) **doubles the bit rate.**

With multilevel signaling (M levels), the Nyquist formula becomes $C = 2\,B \log_2 M$

Note:

Noise and other effects prevent us from increasing the data rate in an arbitrary way using this approach.

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2.6. ”Noise“ („Rauschen“)

The term “noise” refers to **undesired signals which are added** to the original signal somewhere between sender and receiver: Any signal comes along with noise.

The data signal can only be detected correctly, if the **signal-to-noise ratio (SNR)** (“Rauschabstand”) is large enough:

$$\frac{S}{N} = \frac{\text{signal power}}{\text{noise power}} \quad \text{or} \quad \left(\frac{S}{N}\right)^{dB} = 10 \log \left(\frac{S}{N}\right) dB$$

One of the most important categories of noise is **thermal noise** due to thermal agitation of electrons.

Thermal noise is uniformly distributed across the frequency spectrum. It is often called ”**white noise**“.
The thermal noise power may be calculated as follows:

$$N_0 = k\,T \text{ (W/Hz)} \quad \text{resp.} \quad N = k\,T\,B \text{ (W)}$$

N_0: noise power density in Watt/Hz
k: Boltzmann’s constant (1,3803 x 10-23 J/ºK)
T: Temperature, degrees Kelvin (ºC + 273º)
B: channel bandwidth in Hertz

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Maximum Channel Capacity according to Shannon

Shannon‘s result is that the **maximum (error-free !) channel capacity, in bit/s, obeys the equation**

$$C = B \log_2 (1 + S/N)$$

C: max. channel capacity in bit/s
B: channel bandwidth in Hertz
S: signal power
N: noise power



The max. channel bandwidth (in bit/s) increases with the available bandwidth (in Hertz).

Note:

- The noise power also increases with the bandwidth (white noise).
- Shannon’s formula represents the **theoretical maximum**. In practice, only much lower data rates are achieved. Some reasons are
  - Noise other than thermal noise (impulse noise, intermodulation noise, crosstalk)
  - Attenuation and distortion.

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The Expression „E_b / N_o“

The famous expression “E_b / N_o“ is the ratio of

- **energy per bit (E_b)** and
- **noise power density per Hertz (N_0)**

$$\frac{\text{energy per bit [Joule/bit]}}{\text{noise power density [Watt/Hertz]}} = \frac{E_b}{N_0} = \frac{S/R}{kT}$$

S: signal power
R: data rate (bit/s)
N_0: noise power density
k: Boltzmann’s constant
T: temperature in Kelvin

The expression „E_b / N_o“ is important because the **bit error rate** for digital data is a **(decreasing) function of this ratio**.

In radio networks, the “**antenna noise**” often is “converted” to equivalent thermal noise. Thus, “E_b / N_o“ is extensively used for wireless communication.

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Theoretical Bit Error Rates

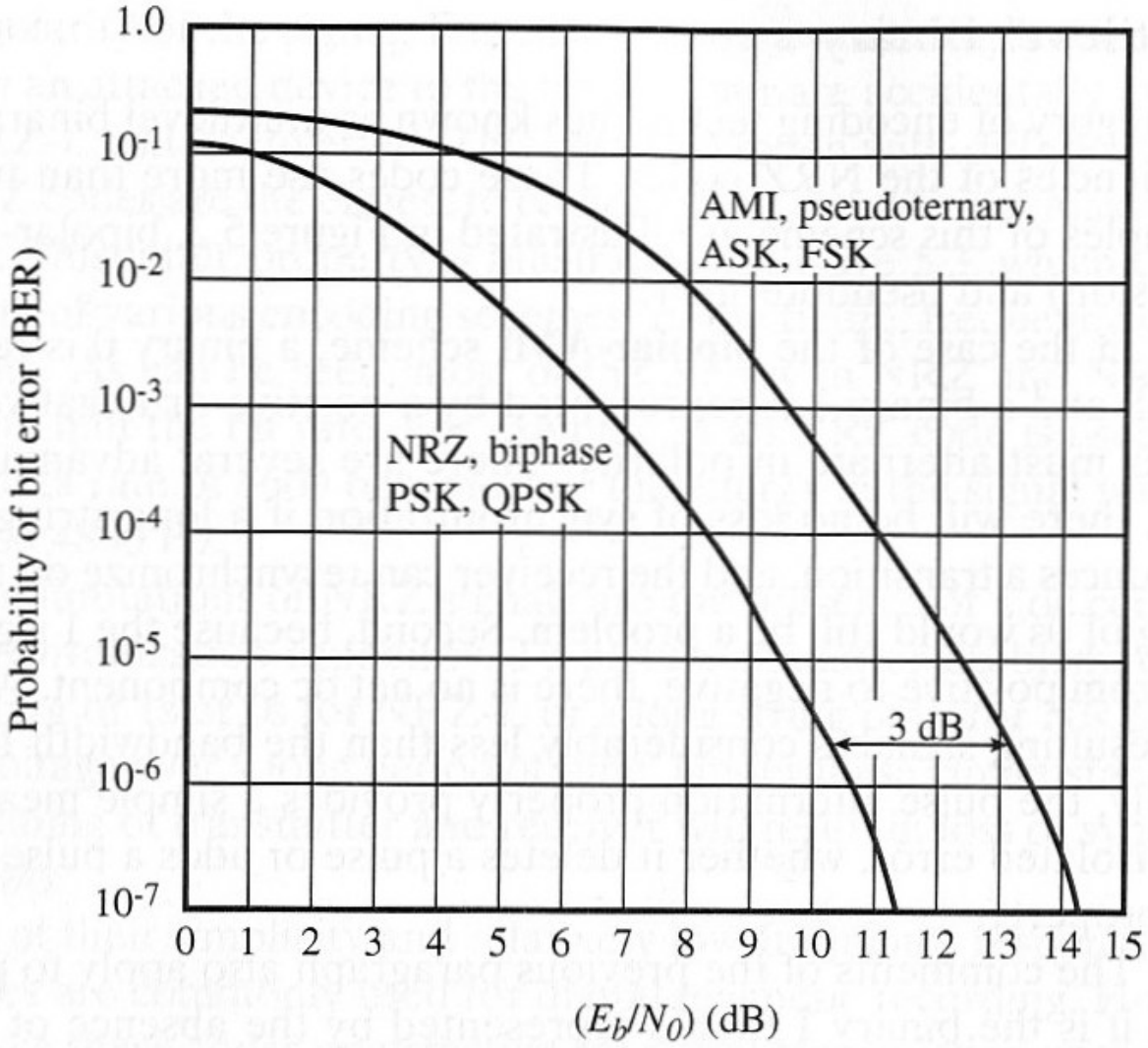


Figure 5.4 Theoretical Bit Error Rate for Various Encoding Schemes

Source: W. Stallings, „Data and Computer Communications“, Prentice Hall, 2000, p. 138

2.7. Analog Transmission of Digital Data

The most familiar case is transmitting digital data across the conventional analog telephone network. The **analog telephone network** was designed to receive, switch, and transmit analog signals in the voice-frequency range of about **300 Hz to 3,400 Hz**.

There are **three basic encoding or modulation techniques** for transforming digital data into analog signals:

- **Amplitude-shift keying (ASK)**
- **Frequency-shift keying (FSK)**
- **Phase-shift keying (PSK and QPSK)**

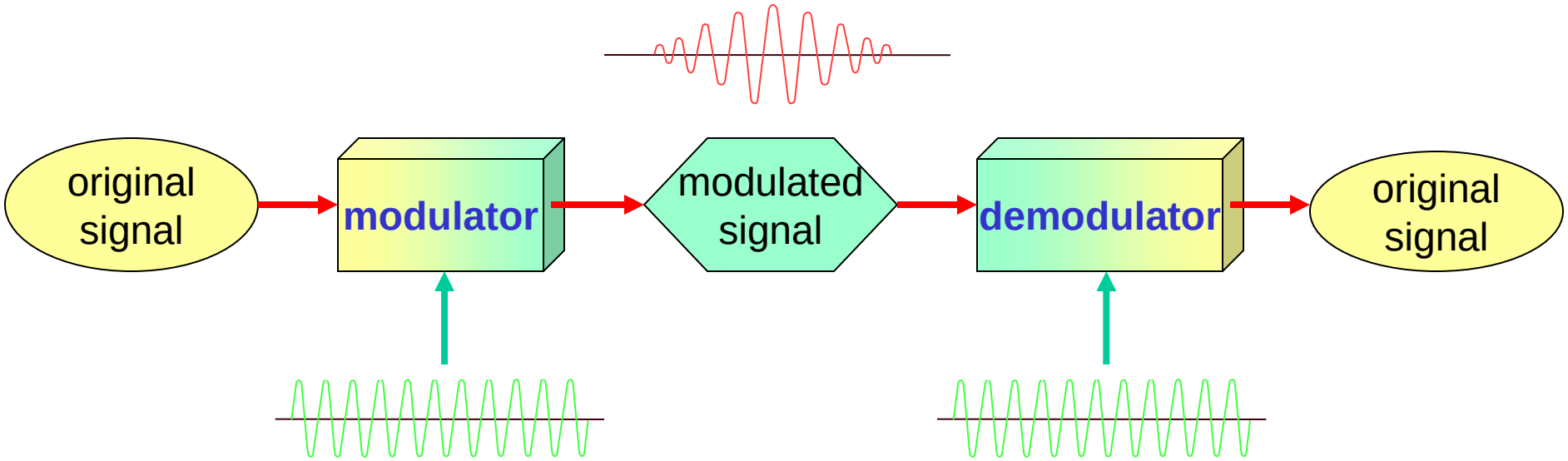
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Basic Concepts

Modulation is
the **variation of a signal parameter** (or: more parameters) **of a "carrier signal"** by a modulating signal.

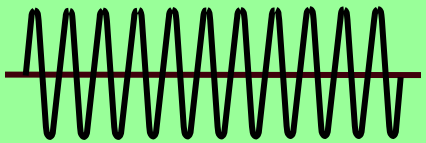
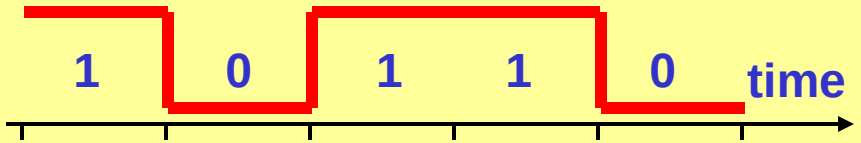
In most cases: Variations of characteristics of an oscillation.

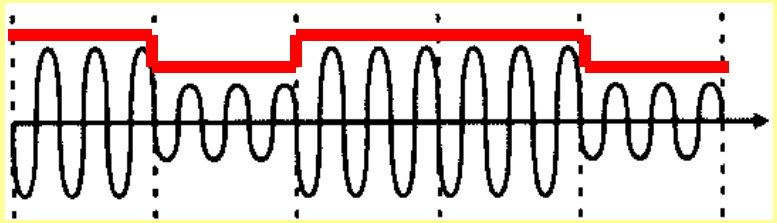
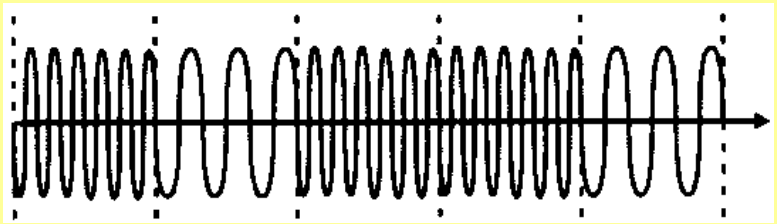
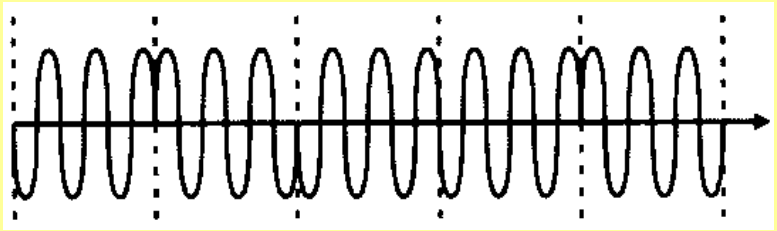
Demodulation is
the **recovery of the original signal** from the modulated signal.



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Overview

carrier signal	original digital signal
	

Modulation of the signal parameter "amplitude"		"0" and "1" are represented by different signal levels (e.g. voltage level).
Modulation of the signal parameter "frequency"		"0" and "1" are represented by different frequencies.
Modulation of the signal parameter "phase"		"0" and "1" are represented by different phasing.

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Amplitude-shift keying (ASK)

In ASK, the two binary values are represented by two different amplitudes of the carrier frequency. Commonly, one of the amplitudes is zero. The resulting signal is:

$$s(t) = \begin{cases} A \cos(2 \pi f_c t) & \text{binary } 1 \\ 0 & \text{binary } 0 \end{cases}$$

f_c the "carrier frequency"

The ASK technique is **used to transmit digital data over optical fiber**. For LED transmitters, the preceding equation is valid. That is,

- **one signal element** is represented by a **light pulse** while
- the **other signal element** is represented by the **absence of light**.

Laser transmitters normally have a fixed "bias" current that causes the device to emit a low light level. This low light level represents one signal element, while a higher-amplitude light wave represents another signal element.

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Frequency-shift keying (FSK)

In FSK, the two binary values are represented by **two different frequencies near the carrier frequency**. The resulting signal is:

$$s(t) = \begin{cases} A \cos(2\pi f_1 t) & \text{binary } 1 \\ A \cos(2\pi f_2 t) & \text{binary } 0 \end{cases}$$

Example: FSK for full-duplex operation over a voice-grade line (300 Hz to 3,400 Hz)

FSK in one direction centered on 1,170 Hz with 100 Hz shift on either side.
Similarly - for the other direction – centered on 2,125 Hz.

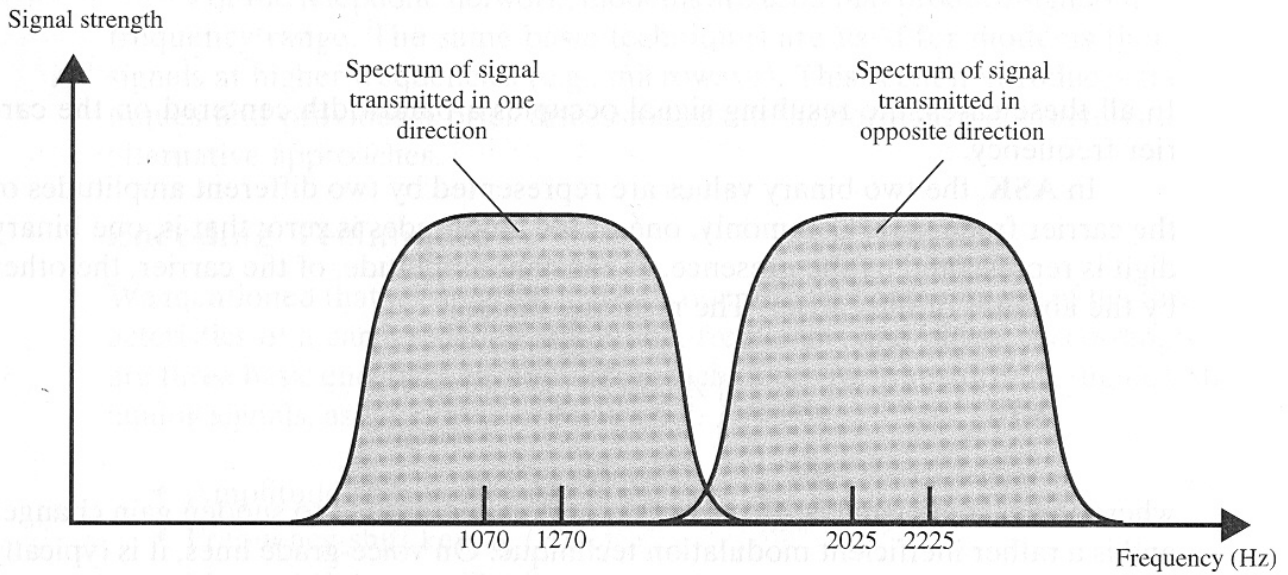


Figure 5.8 Full-Duplex FSK Transmission on a Voice-Grade Line

Source: W. Stallings, "Data and Computer Communications", Prentice Hall, 2000, p. 144

Phase-shift keying (PSK and QPSK)

In PSK, the phase of the carrier signal is shifted to represent data:

$$s(t) = \begin{cases} A \cos(2\pi f_c t + \pi) & \text{binary } 1 \\ A \cos(2\pi f_c t) & \text{binary } 0 \end{cases}$$

The mechanism shown here is called **"differential PSK"**, because the phase shift is with reference to the previous bit transmitted rather than to some constant reference signal.

More efficient use of bandwidth can be achieved if each signaling element represents more than one bit. For example, instead of a phase shift of 180º, as allowed in PSK, a common encoding technique, known as **"quadrature phase-shift keying" (QPSK)**, uses phase shifts of multiples of 90º:

$$s(t) = \begin{cases} A \cos(2\pi f_c t + \frac{\pi}{4}) & \text{for } 11 \\ A \cos(2\pi f_c t + \frac{3\pi}{4}) & \text{for } 10 \\ A \cos(2\pi f_c t + \frac{5\pi}{4}) & \text{for } 00 \\ A \cos(2\pi f_c t + \frac{7\pi}{4}) & \text{for } 01 \end{cases}$$

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Quadrature Amplitude Modulation (QAM)

QAM is a widely (e.g. in DSL) used modulation scheme which conveys data by modulating the amplitude of two carrier waves.

QAM is a combination of phase and amplitude modulation.

The figure illustrates the idea.

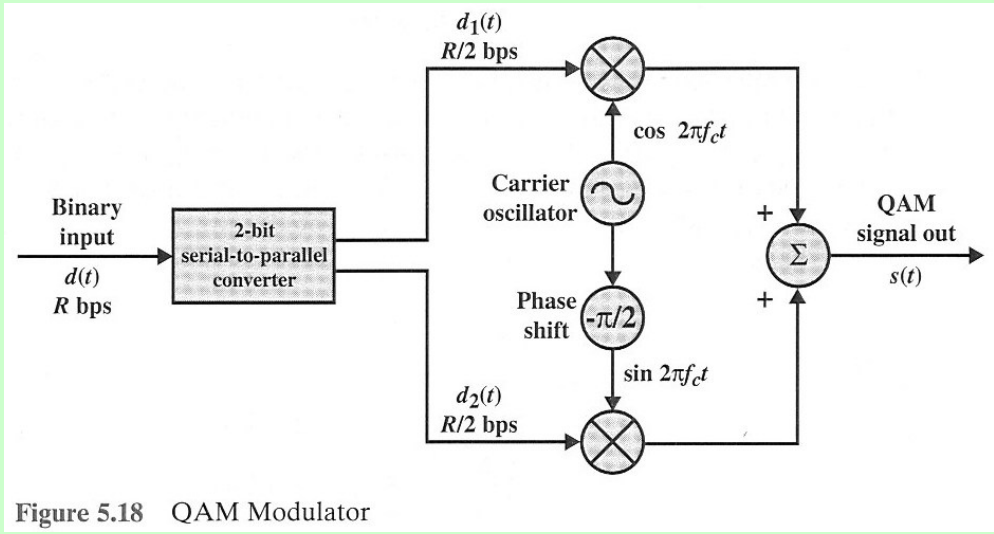


Figure 5.18 QAM Modulator

Source: W. Stallings, „Data and Computer Communications“, Prentice Hall, 2000, p. 144

The transmitted signal may be expressed as:

$$s(t) = d_1(t) \cos(2\pi f_c t) + d_2(t) \sin(2\pi f_c t)$$

With **two-level ASK** each of the 2 data streams can be in one of two states.

The **combined stream can be in one of 4 states** (2 x 2).

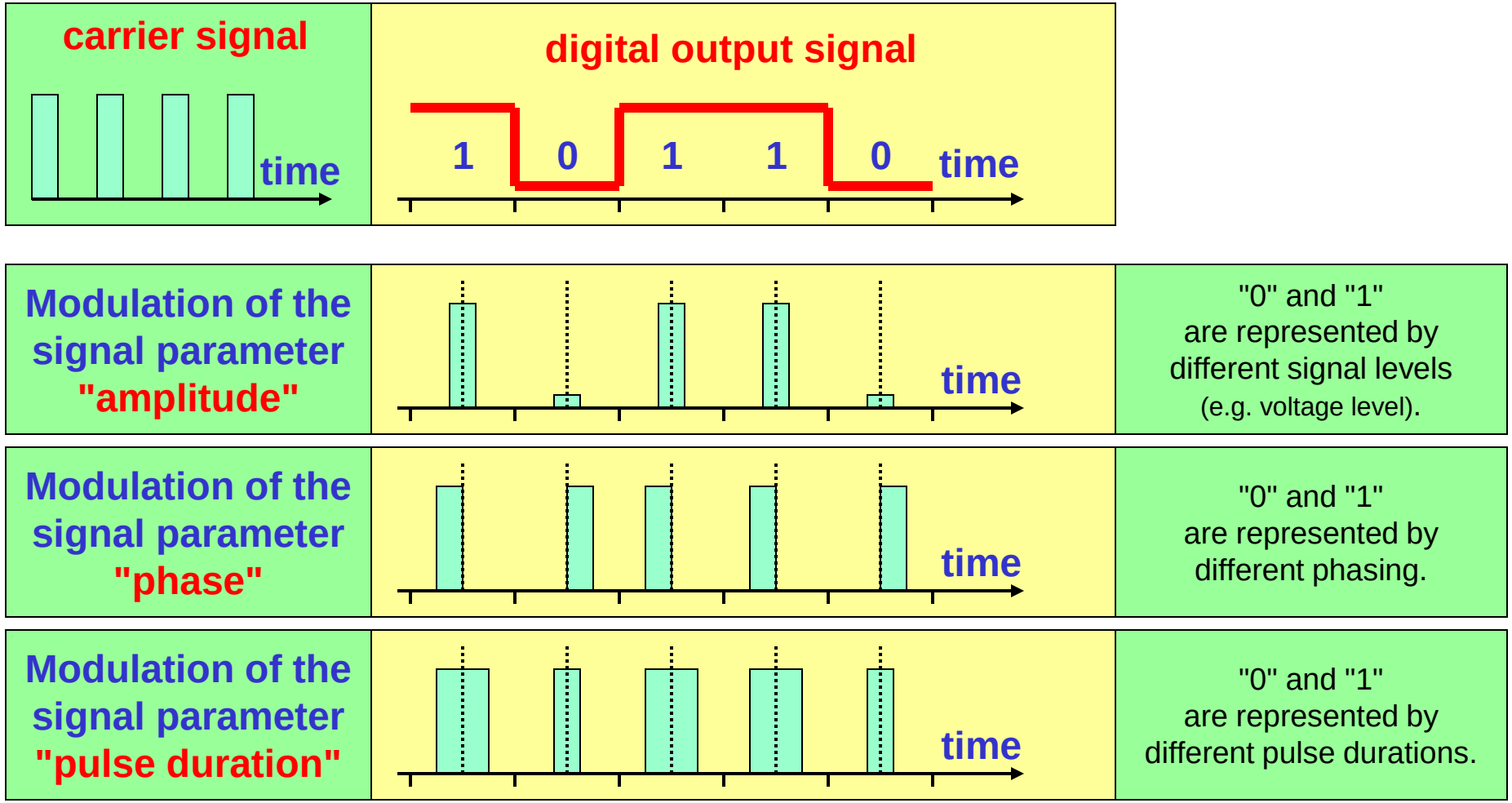
With **4-level ASK** the **combined stream can be in one of 16 states** (4 x 4).

Systems with 64 states, 256 states (and more) have been implemented.

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2.8. Transmission of Digital Data with Pulse Code Modulation

Even when using a digital carrier, different signal parameters can be modified to represent the information.



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2.9. Communication requires Synchronization

When using time discrete signals, the receiver must be aware of the **timing**:



Receiver and **sender have to be synchronized**.

2.9.1. "NRZ" (No-Return-to-Zero)

2.9.2. Asynchronous Communication

2.9.3. Synchronous Communication

2.9.4. Manchester Code

2.9.5. Differential Code (NRZI)

2.9.6. Selection of Adequate Code Words

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2.9.1. "NRZ" (No-Return-to-Zero)

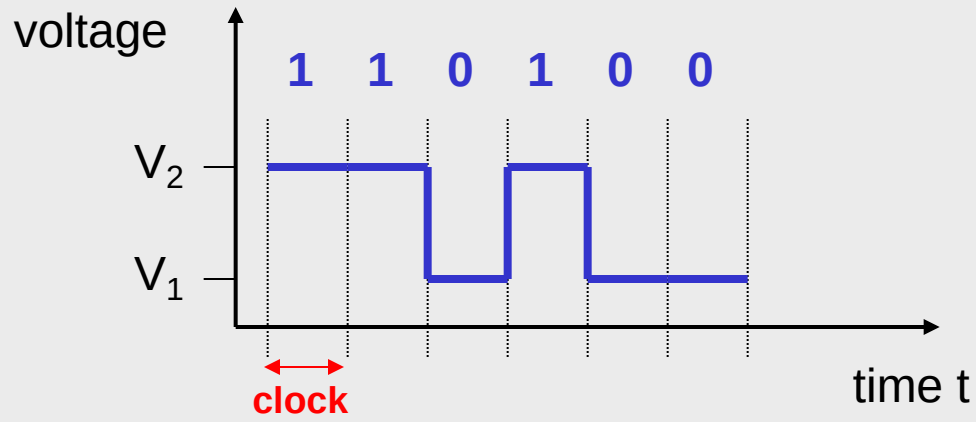
At first glance, representation of binary data seems to be trivial:

➡ Assign **fixed values** (of a signal parameter) to "0" and "1".

Examples:	"0"	0 volts	"light off"
	"1"	+5 volts	"light on"

This kind of signal representation is called **"NRZ" (No-Return-to-Zero)**:

Example:



A shorter clock cycle results in **a higher data rate**:

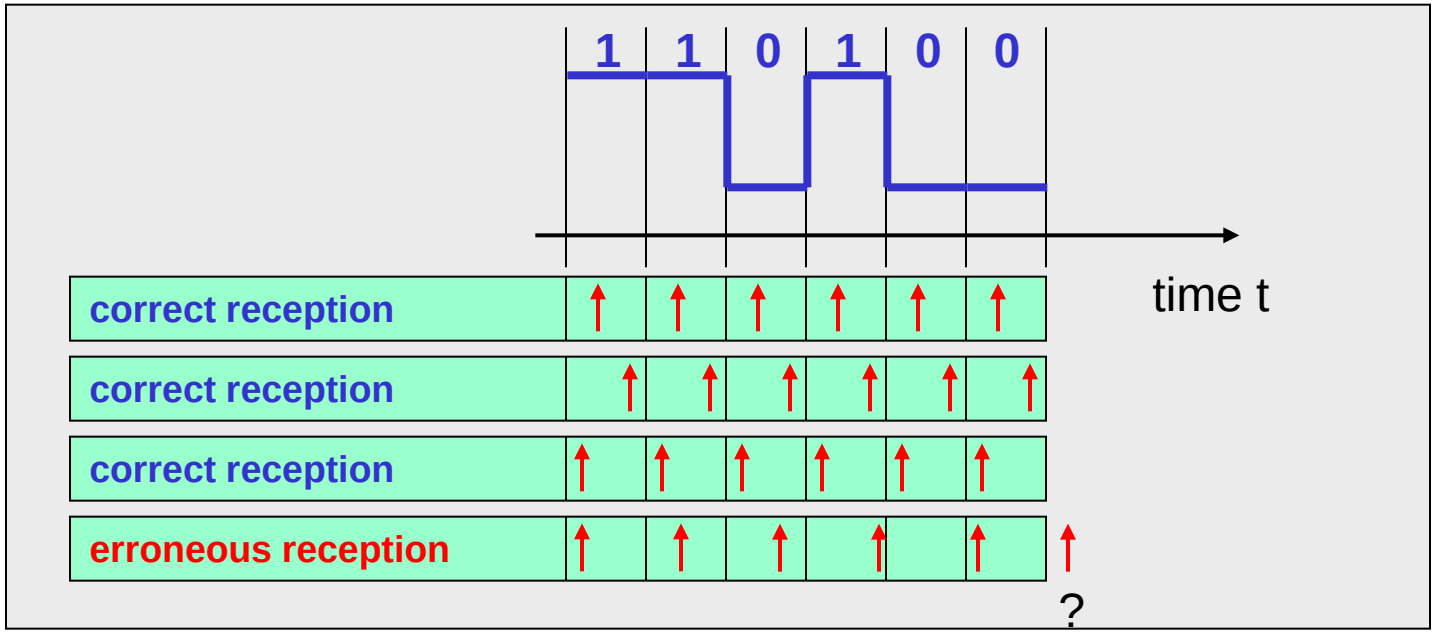
➡ 1 million clock cycles/s allow for a data rate of 1Mbit/s*

*Attention: in the world of networking, in most cases 1 M = 1,000,000; 1 k = 1,000.

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Receiving a NRZ signal

The receiver of a NRZ signal has to **check the signal level in each interval** in order to decode the binary value. ... But this **may be tricky** ...



Observations:

- The error-free reception of a message works only if the receiver samples the signal at **the correct points in time** (e.g. in the middle of a bit.)
- The synchronization of the receiver to the sender is trivial if the receiver can adapt to **signal level changes**.
- With **NRZ**, extensive sequences of ones or zeros cause **synchronization problems**.

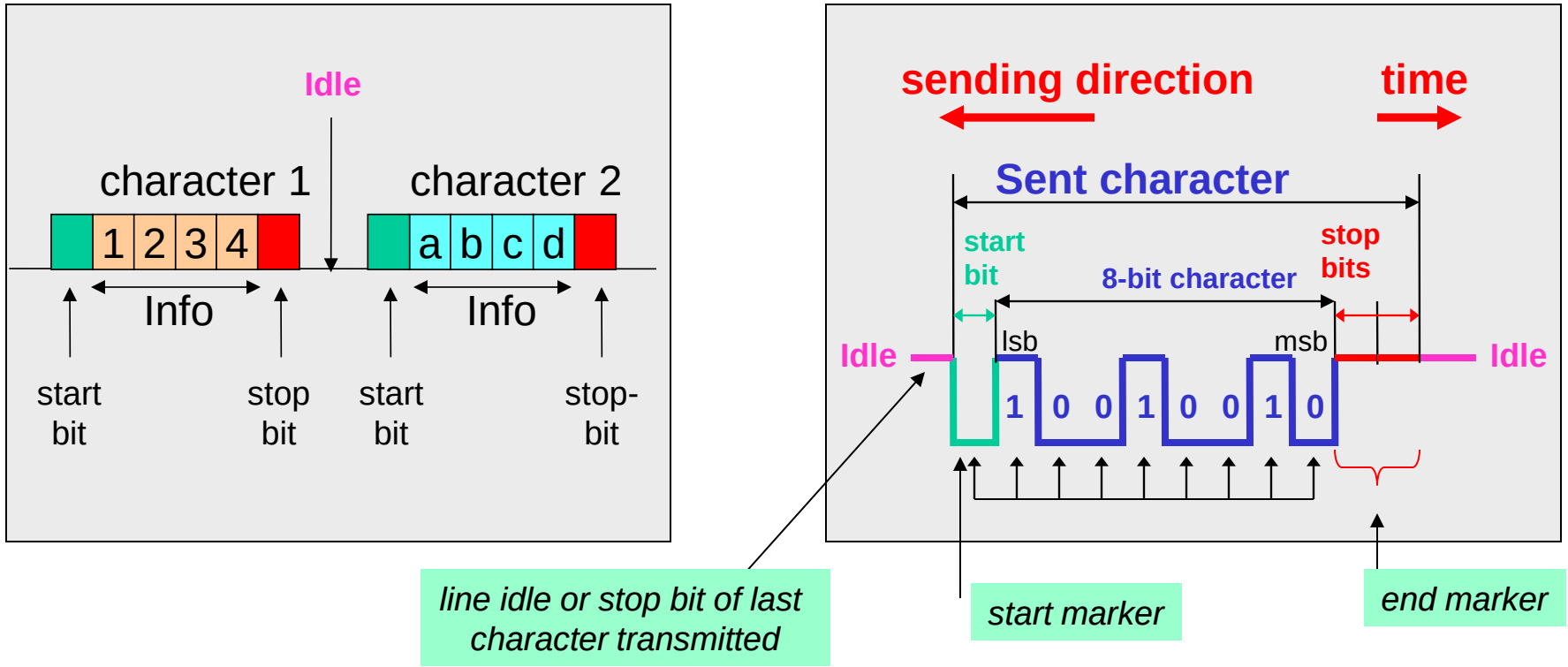
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2.9.2. Asynchronous Communication

Asynchronous communication is the "classical" way of transmitting ASCII characters.

Idea: - **"wrap"** the "characters" with **start-** and **stop-bits** (or start-/stop-characters)

- This guarantees **adequate synchronization**.



1, 1.5 or 2 stop-bits mark the end of each character.

Stop-bit signal level is different from start-bit signal level. This guarantees start bit detection.

2.9.3. Synchronous Communication

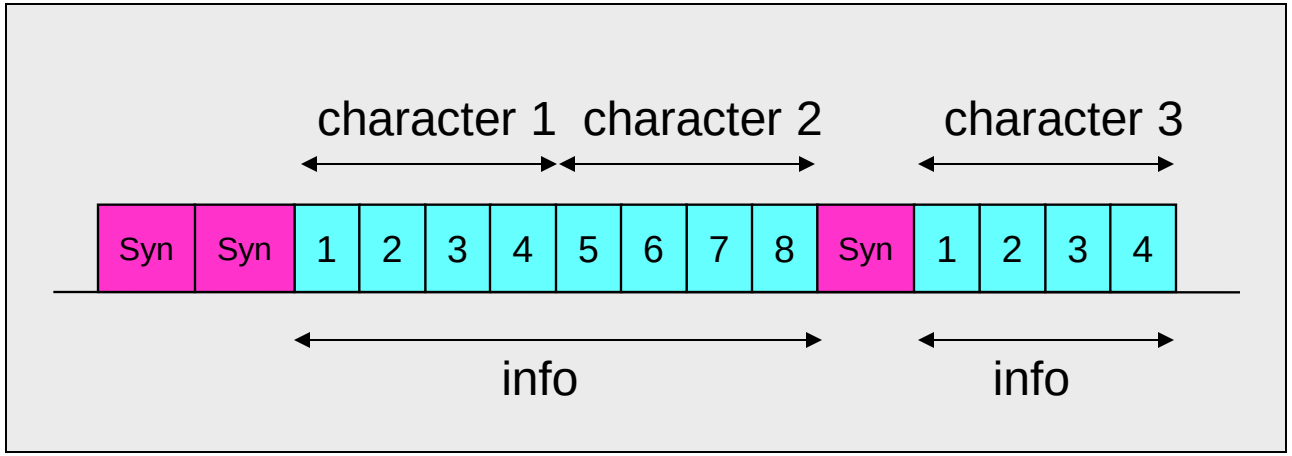
The receiver is **always synchronized to the sender**.

a) During message transmission

→ synchronization derived from the **signal received**

b) Otherwise

→ Send **special "sync" signals**



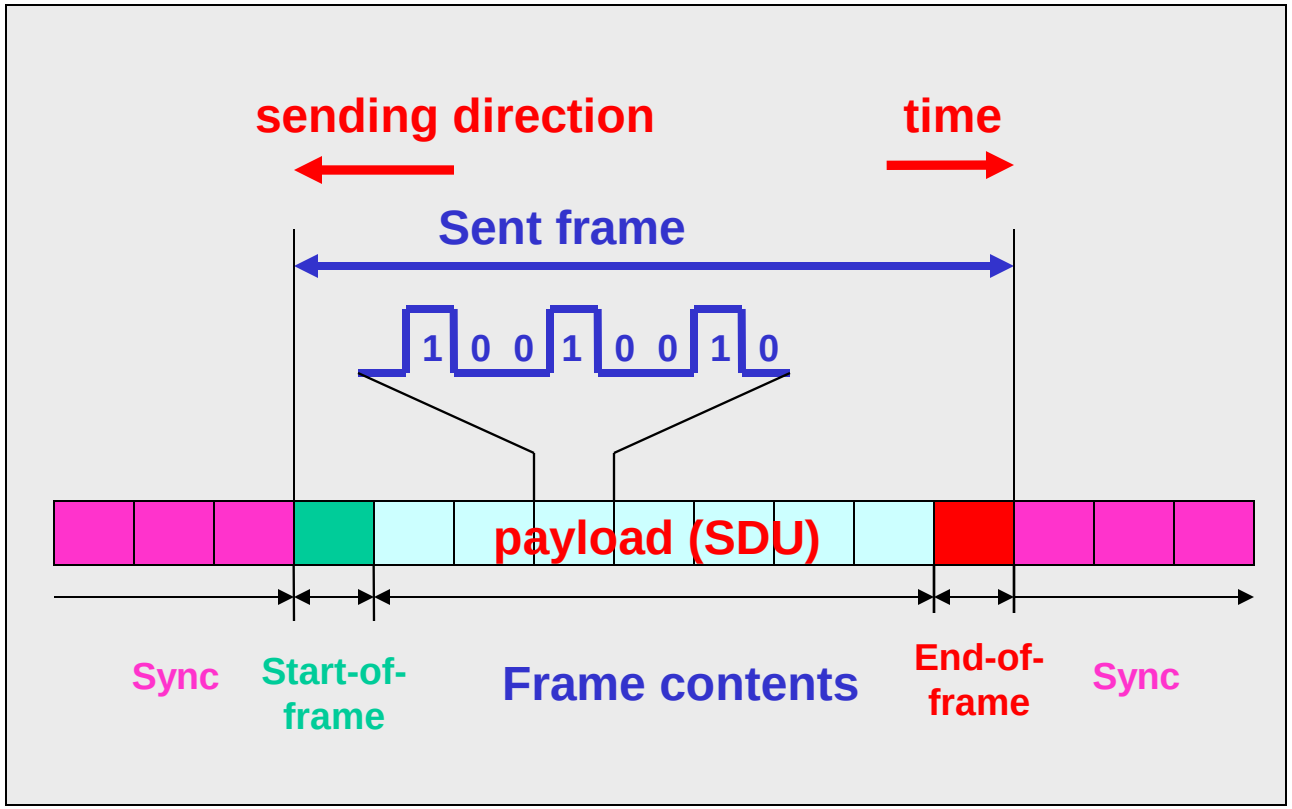
Obviously, **sync signals need to be distinguished from "ordinary" signals**.

Close interaction of OSI layers 1 and 2 helps:

Layer 2 defines a framework for the structuring of a bit stream.

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Synchronous Communication (2)



A "frame" is a **layer 2 PDU** (OSI model).

Layer 1 may include **special "codes" for marking start and stop**.

Alternatively, **"start-of-frame"** and **"end-of-frame"** may be defined as **unambiguous bit strings**. (discussed later)

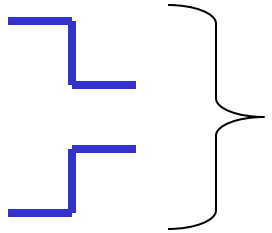
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2.9.4. Manchester Code

In the so-called "Manchester code", e.g. used in classic Ethernet, there always is a **transition in the middle of the bit**:

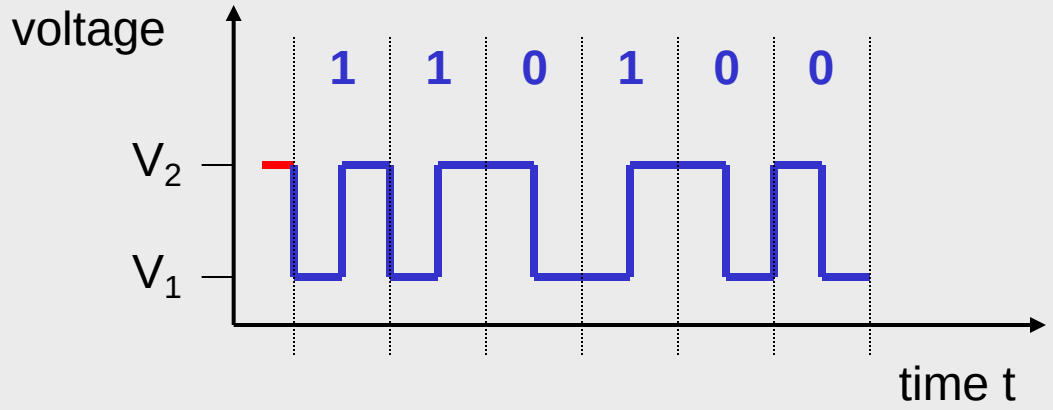
"0" = 'High' in 1st half, 'Low' in 2nd half

"1" = 'Low' in 1st half, 'High' in 2nd half



Similarly: inverse version

Example:



Advantage:

- **Self-clocking**
- **No problems with sequences** of zeros or ones.
- **End of data transmission is easy to recognize** (no more transitions).

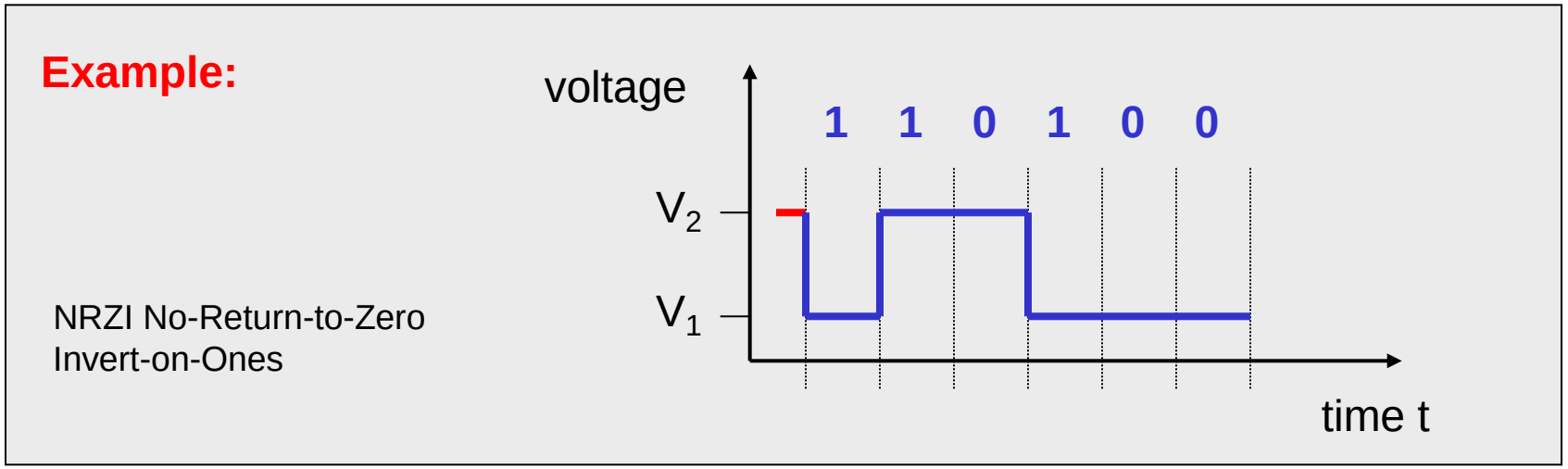
Disadvantage: **Two characteristic units** per bit (double clock rate or half data rate).

2.9.5. Differential Code (NRZI)

"0": no transition
"1": always transition

NRZI: No-Return-to-Zero Invert-on-Ones

Example:



Advantage: One bit per clock cycle.

Disadvantage: No transitions during sequences of zeros.

Synchronization of receiver to sender becomes trivial if **max. distance between ones is small enough**.

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2.9.6. Selection of Adequate Code Words

Codes with a **"clever" choice of code words** are used in different systems.

An example for line coding with selection of specific code words is the so-called **"(4B,5B)-code"**, used with the **"Fiber Distributed Data Interface"** (FDDI).

The (4B,5B)-code

- codes **4 bit with 5 bit**
- uses **NRZI-code** (transition with "1")
- uses the **most useful bit strings** from the set of 32 possible strings
  (lots of ones, i.e. lots of transitions)

Compromise: efficiency $\leftrightarrow$ self-clocking

Code Word Selection is also extensively used in high speed networks like 100 Mbit/s Ethernet to reduce the bandwidth required.

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(4B,5B)-Code Symbol Representation

Decimal	Code Group	Assignment
Line State Symbols		
00	00000	QUIET
31	11111	IDLE ("Sync")
04	00100	HALT
Starting Delimiter		
24	11000	1st of Sequential SD Pair
17	10001	2nd of Sequential SD Pair
Ending Delimiter		
13	01101	Terminate Data Stream
Control Indicators		
07	00111	Logical ZERO (Reset)
25	11001	Logical One (Set)
Invalid Code Assignments		
01	00001	
02	00010	
03	00011	
05	00101	
06	00110	
08	01000	
12	01100	
16	10000	

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(4B,5B)-Code Symbol Representation (2)

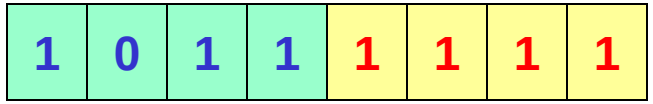
Decimal	Code Group	Assignment	
Data Symbols		Binary	Hex
30	11110	0000	0
09	01001	0001	1
20	10100	0010	2
21	10101	0011	3
10	01010	0100	4
11	01011	0101	5
14	01110	0110	6
15	01111	0111	7
18	10010	1000	8
19	10011	1001	9
22	10110	1010	A
23	10111	1011	B
26	11010	1100	C
27	11011	1101	D
28	11100	1110	E
29	11101	1111	F

There are at least two transitions in each code group. There is no transmission of more than three bits without transition (three zeros back-to-back).

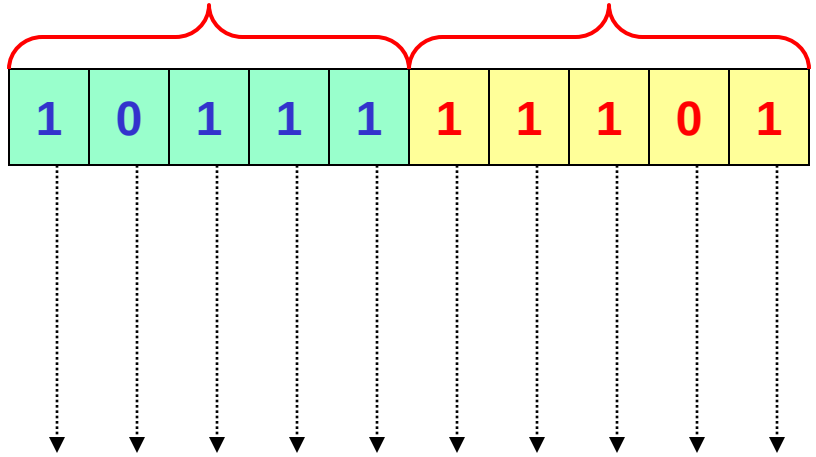
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Example: Usage of (4B,5B)-Codes

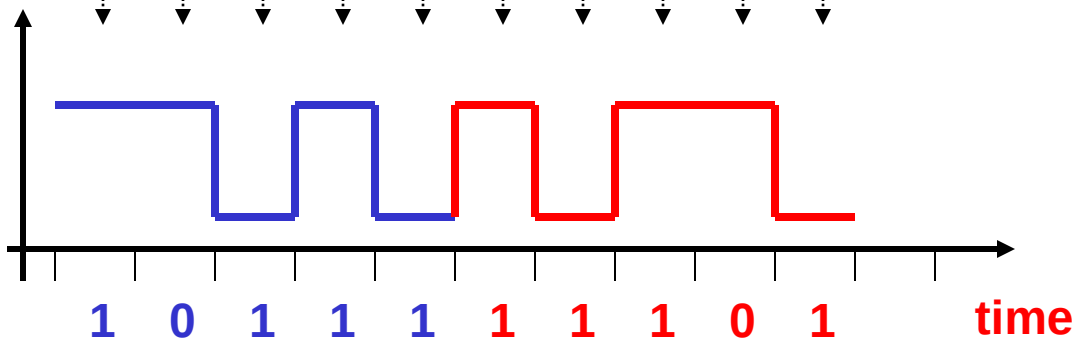
To transmit:



Encode 4 bits by 5 bits:



Transmit according to NRZI:



(8B6T)-Coding as specified for 100 Mbit/s Ethernet

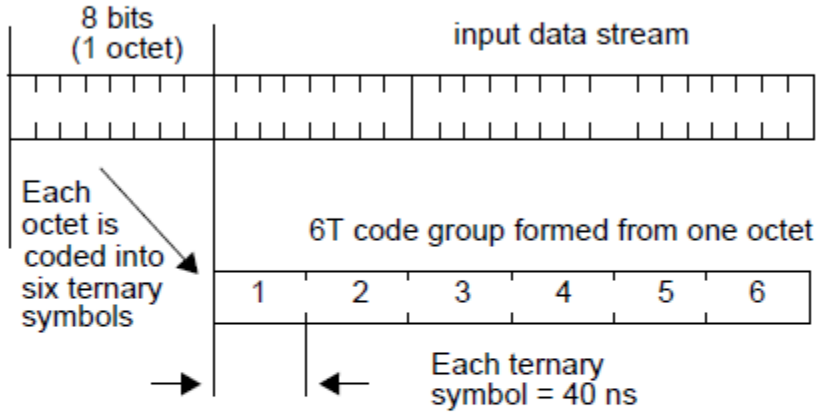


Figure 23–4—8B6T coding

8B6T coding, as used with 100BASE-T4 signaling, maps data octets into ternary symbols. Each octet is mapped to a pattern of 6 ternary symbols, called a 6T code group. The 6T code groups are fanned out to three independent serial channels. The effective data rate carried on each pair is one third of 100 Mb/s, which is 33.333... Mb/s. The ternary symbol transmission rate on each pair is 6/8 times 33.33 Mb/s, or precisely 25.000 MHz.

http://standards.ieee.org/getieee802/download/802.3-2012.zip

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The 6T Code specified for 100 MBit/s Ethernet

Annex 23A

(normative)

6T codewords

The *leftmost* ternary symbol of each 6T Code group shown in Table 23A–1 (broken into 23A–1a and 23A–1b for pagination) shall be transmitted *first*. The *leftmost* nibble of each data octet is the *most significant*.

Table 23A–1a—100BASE-T4 8B6T code table

Data octet	6T code group	Data octet	6T code group	Data octet	6T code group	Data octet	6T code group
00	+ - 0 0 + -	20	0 0 - + + -	40	+ 0 + 0 0 -	60	0 - 0 + + 0
01	0 + - + - 0	21	- - + 0 0 +	41	+ + 0 0 - 0	61	0 0 - + 0 +
02	+ - 0 + - 0	22	+ + - 0 + -	42	+ 0 + 0 - 0	62	0 - 0 + 0 +
03	- 0 + + - 0	23	+ + - 0 - +	43	0 + + 0 - 0	63	- 0 0 + 0 +
04	- 0 + 0 + -	24	0 0 + 0 - +	44	0 + + 0 0 -	64	- 0 0 + + 0
05	0 + - - 0 +	25	0 0 + 0 + -	45	+ + 0 - 0 0	65	0 0 - 0 + +
06	+ - 0 - 0 +	26	0 0 - 0 0 +	46	+ 0 + - 0 0	66	0 - 0 0 + +
07	- 0 + - 0 +	27	- - + + + -	47	0 + + - 0 0	67	- 0 0 0 + +
08	- + 0 0 + -	28	- 0 - + + 0	48	0 0 0 + 0 0	68	- + - + + 0
09	0 - + + - 0	29	- - 0 + 0 +	49	0 0 0 - + +	69	- - + + 0 +
0A	- + 0 + - 0	2A	- 0 - + 0 +	4A	0 0 0 + - +	6A	- + - + 0 +
0B	+ 0 - + - 0	2B	0 - - + 0 +	4B	0 0 0 + + -	6B	+ - - + 0 +
0C	+ 0 - 0 + -	2C	0 - - + + 0	4C	0 0 0 - + 0	6C	+ - - + + 0
0D	0 - + - 0 +	2D	- - 0 0 + +	4D	0 0 0 - 0 +	6D	- - + 0 + +
0E	- + 0 - 0 +	2E	- 0 - 0 + +	4E	0 0 0 + - 0	6E	- + - 0 + +
0F	+ 0 - - 0 +	2F	0 - - 0 + +	4F	0 0 0 + 0 -	6F	+ - - 0 + +
10	+ 0 + - - 0	30	+ - 0 0 - +	50	+ 0 + - - +	70	- + + 0 0 0
11	+ + 0 - 0 -	31	0 + - - + 0	51	+ + 0 - + -	71	+ - + 0 0 0
12	+ 0 + - 0 -	32	+ - 0 - + 0	52	+ 0 + - + -	72	+ + - 0 0 0
13	0 + + - 0 -	33	- 0 + - + 0	53	0 + + - + -	73	0 0 + 0 0 0
14	0 + + - - 0	34	- 0 + 0 - +	54	0 + + - - +	74	- 0 + 0 0 0
15	+ + 0 0 - -	35	0 + - + 0 -	55	+ + 0 + - -	75	0 - + 0 0 0
16	+ 0 + 0 - -	36	+ - 0 + 0 -	56	+ 0 + + - -	76	+ 0 - 0 0 0
17	0 + + 0 - -	37	- 0 + + 0 -	57	0 + + + - -	77	0 + - 0 0 0
18	0 + - 0 + -	38	- + 0 0 - +	58	+ + + 0 - -	78	0 - - + + +
19	0 + - 0 - +	39	0 - + - + 0	59	+ + + - 0 -	79	- 0 - + + +
1A	0 + - + + -	3A	- + 0 - + 0	5A	+ + + - - 0	7A	- - 0 + + +
1B	0 + - 0 0 +	3B	+ 0 - - + 0	5B	+ + 0 - - 0	7B	- - 0 + + 0
1C	0 - + 0 0 +	3C	+ 0 - 0 - +	5C	+ + 0 - - +	7C	+ + - 0 0 -
1D	0 - + + + -	3D	0 - + + 0 -	5D	+ + 0 0 0 -	7D	0 0 + 0 0 -
1E	0 - + 0 - +	3E	- + 0 + 0 -	5E	- - + + + 0	7E	+ + - - - +
1F	0 - + 0 + -	3F	+ 0 - + 0 -	5F	0 0 - + + 0	7F	0 0 + - - +